Posting Like a State: How do Citizens Interact with Autocrats' Social Media

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Abstract

This study examines strategic political communication under authoritarian rule through official social media, an infrastructure of state–society interaction. Using 4.21 million posts from 329 Chinese local-government WeChat accounts, I show that official accounts are mixed channels that provide public-service information and propaganda. First, I document a communication dilemma: service content sustains attention but draws higher readership than propaganda, crowding propaganda out in the same channel. I then use a regression discontinuity design around two platform reforms to show that interface design shapes politically visible responses. When WeChat made expressions of approval publicly visible in December 2018, interaction fell by 1.9 percentage points; when it restored a less visible way to register approval in July 2020, interaction increased by 2.9 points. Visible-endorsement avoidance is strongest for hard propaganda and higher for soft propaganda than for public-service content, suggesting reverse preference falsification in state–society interaction through digital media.

Keywords: Digital Authoritarianism; Preference Falsification; Propaganda; Social Media; State–Society Interaction.

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1 Introduction

How do citizens in autocracies interact with official social media, and what do those traces actually mean? As propaganda, public communication, and service delivery migrate onto digital platforms, state-controlled accounts accumulate vast numbers of reads, likes, and shares. These metrics are tempting to interpret as indicators of propaganda success or regime support. Yet in authoritarian settings, interacting with official content is rarely a purely expressive act. Citizens may engage because they need information, because a post is difficult to avoid, because approval is low-cost, or because visible endorsement carries reputational and political risk. The political meaning of digital engagement is therefore fundamentally ambiguous.

Existing work speaks to pieces of this puzzle, but rarely to the mechanism at stake here. Research on Chinese propaganda shows that the party-state competes for attention through commercialization, soft propaganda, distraction, and popularization rather than relying on crude indoctrination alone (Brady 2008; King, Pan, and Roberts 2017; Lu and Pan 2021; Mattingly and Yao 2022; Repnikova and Fang 2018; Roberts 2018; Stockmann 2013; Wang 2024; Zou 2021). Research on preference falsification and self-censorship emphasizes the gap between private belief and public expression under pressure (Carter, Carter, and Schick 2024; Jiang and Yang 2016; Kuran 1995; Noelle-Neumann 1993; Shen and Truex 2020). A third literature shows that government WeChat accounts are not merely propaganda outlets: local authorities also use them for public service delivery, routine contact, and audience acquisition (Jiang et al. 2021; Liu, Ma, and Xia 2024; Yu and Shen 2026; Zheng 2013; Zheng and Zheng 2014). A related strand highlights semi-official accounts, delegated censorship, and layered platform governance, suggesting that state influence is often embedded in organizationally ambiguous or platform-native forms (Neagli 2021; Sun and Zhao 2022; Tai and Fu 2020). What these literatures do not yet explain well is how platform design changes the political meaning of interaction itself.

I argue that official social media in digital authoritarianism are mixed channels that bundle public service information with political messaging, and that citizens sort their behavior across different levels of expressive risk. On WeChat, reading is low-risk consumption, a like is low-

visibility approval, *zaikan* is socially legible endorsement, and sharing is the highest-visibility form of amplification. This layered-expression perspective shifts the theoretical question away from whether citizens "believe" propaganda and toward how they manage observable traces of alignment.

I evaluate this argument using 4.21 million post titles published by 329 prefecture-level government WeChat official accounts between 2015 and 2024. The article makes two empirical moves. First, I use a theory-driven 'LLM + human' classification workflow to assign fine-grained post labels and then collapse them into three main content families: hard propaganda, soft propaganda, and public service. Second, I use the 2018 and 2020 WeChat interface reforms to motivate a redesigned empirical analysis of how platform changes alter the meaning of a click.

This article makes three contributions. First, it extends preference falsification theory from speech to platform-mediated behavior by showing that digital expression is layered across reading, liking, *zaikan*, and sharing. Second, it contributes to the study of propaganda and media consumption by separating attention from endorsement and by showing how service content both builds and competes for political attention. Third, it recasts official social media as an everyday site of authoritarian state–society interaction, where governments manage not only information but also the terms under which citizens can safely engage.

2 Theory and Hypotheses

Official social media in authoritarian settings should be understood as a platformed interaction regime rather than as a simple channel of state persuasion. In classic preference falsification accounts, the central distinction is between private preference and public statement. On platforms such as WeChat, however, expression is not binary. The interface decomposes response into multiple actions that expose users to different audiences and therefore to different expressive risks. The state may observe aggregate traces, but friends, colleagues, and acquaintances can also infer meaning from visible acts. Digital preference falsification is therefore layered: the same user may

privately consume official content, register low-visibility approval, and avoid socially visible endorsement (Kuran 1995; Noelle-Neumann 1993; Shen and Truex 2020; Tsai and Men 2018; Xu 2022).

This layered-expression logic matters because Chinese government WeChat accounts are mixed channels rather than purely political outlets. Existing research shows that state agencies use social media not only to disseminate political messages but also to provide services, route transactions, manage crises, and sustain routine interaction with the public (Jiang et al. 2021; Liu, Ma, and Xia 2024; Yu and Shen 2026; Zheng 2013; Zheng and Zheng 2014). That literature is especially useful because it shows that many users approach official accounts as lookup tools or service interfaces rather than as ideological venues. Service content therefore helps governments build durable audiences, normalize repeated contact, and increase the probability that users encounter future political content. At the moment of consumption, however, service information also competes with overt political messaging for scarce attention. This is the paper's attention dilemma.

The same logic also changes how we should think about propaganda. Contemporary influence online often works through participatory persuasion, semi-official intermediaries, and delegated forms of control rather than through a single visible propaganda apparatus (King, Pan, and Roberts 2017; Lu and Pan 2021; Neagli 2021; Repnikova and Fang 2018; Sun and Zhao 2022). Hard propaganda foregrounds hierarchy, ideology, discipline, and coercive state capacity. Soft propaganda packages regime-favorable messages in more appealing, emotionally resonant, or developmental frames (Mattingly and Yao 2022; Pan, Shao, and Xu 2022; Wang 2024). Citizens may read all three kinds of content, but the willingness to leave an observable trace should vary with both content risk and interface design. Platform reforms do not merely change aesthetics. They reprice the political and social costs of interacting with the state.

This framework yields four expectations:

H1 (Attention and Utility). Public service content should attract more routine consumption than either hard or soft propaganda because it provides immediate utility and lower expressive stakes.

H2 (Rebundling Visible Approval). When the 2018 redesign bundles approval with public rec-

ommendation, low-visibility like responses should collapse and observed reactions should be reallocated into more socially legible channels.

H3 (Private Substitution). When low-visibility likes return in 2020, approval should shift back toward likes rather than rising uniformly across all public-facing reactions.

H4 (Political-Risk Gradient). These reallocations should be strongest for hard propaganda, then soft propaganda, and weakest for routine public-service content.

Table 1: Platform reforms and theoretical expectations

Time point	Brief policy content	Reading / exposure	Visible endorsement	Low-risk approval
2017–05 (v6.5.8 rollout)	WeChat introduces "See" (*kanyikan*), adding a friend- and algorithm-based discovery channel while leaving the bottom "like" button unchanged.	+	0	0
2018–12–21	WeChat 7.0 replaces "like" with "good-looking" (*haokan*, later *zaikan*), bundling approval with recommendation into "See" (*kanyikan*).	0	reallocated / ambiguous	–
2020–06–29	The "like" button is restored at the bottom of public-account articles alongside the existing "seen" (*zaikan*) and share buttons.	0	0/–	+

Note: '+' indicates an expected increase, '-' an expected decrease, and '0' little immediate change. The 2018 redesign mechanically rebundles approval with recommendation, so observed public-signal measures may be reallocated rather than move with a single clean sign. The strongest directional effects are expected for hard propaganda.

3 Setting, Data, and Measurement

WeChat is the central digital infrastructure of everyday life in China. Since its launch in 2011, it has evolved into a multipurpose "super app" that combines messaging, payments, news distribution, and social networking within a single platform. By the 2020s, it had become the primary interface through which Chinese citizens communicate, consume news, and access digital public and private services. For a study of official digital communication in China, WeChat matters not because it is just another social media platform, but because it is deeply embedded in ordinary social, administrative, and economic life (Ng 2015; Roberts 2020).

Within this ecosystem, "public accounts" operate as mixed institutional channels. They transmit propaganda and political messaging, publicize leadership priorities, and provide highly practical information and service announcements (Jiang et al. 2021; Liu, Ma, and Xia 2024; Yu and Shen 2026; Zheng 2013; Zheng and Zheng 2014). At the same time, the broader WeChat ecosystem includes semi-official and organizationally ambiguous accounts that can carry state-aligned messages without looking fully official (Neagli 2021). A typical article displays a read counter together with several engagement mechanisms: a like button, *haokan*/*zaikan*, sharing, and collecting. These buttons do not measure the same thing with different labels. They ask users to leave different kinds of traces under different conditions of visibility.

The key institutional changes are the 2018 and 2020 interface reforms. In December 2018, WeChat replaced the original like button on newly pushed public-account articles with a "good-looking" button tightly integrated with the "See" tab. A previously quasi-private like was transformed into a socially legible recommendation. In June 2020, WeChat restored the traditional like button alongside *zaikan* and sharing, thereby unbundling low-visibility approval from public recommendation. I treat 2017 as background discovery reform and 2018/2020 as the two main discontinuities for expressive engagement. Table 1 summarizes these shifts.

This study uses the universe of posts published by 329 prefecture-level government WeChat public accounts between 1 January 2015 and 31 December 2024, covering 31 provincial-level units and 4,207,072 posts. The unit of observation for classification is the individual post title. For each post, I record the account identifier, publication date, title, and engagement indicators displayed in the interface, including reads, likes, shares, collecting, and *haokan*/*zaikan*. The archived source file also preserves a first-pass category field, a short reason field, a keyword field, and a confidence score from an earlier coarse classification pass; these metadata are used only for title-based normalization and coding, not as outcomes. Because WeChat top-codes some counters at 100,000 (displayed as "100k+"), I recode those values to 100,001 so they remain ordered while preserving censoring at the upper bound.

To categorize post content at scale, I use an LLM as a text coder rather than as a generator.

The production prompt is set before full-corpus assignment and uses only title-side inputs: title, account name, province, city, publication date, keywords, and the short reason field. Engagement outcomes, account-level outcome patterns, and reform-period indicators never enter the classification prompt. The coding workflow is built on a fixed ten-category taxonomy. The main text relies on this workflow only for measurement credibility; the SI documents taxonomy construction, coder-audit rules, and validation diagnostics.

I validate the LLM labels against a dual-coded gold-standard sample of 2,000 post titles coded under the same codebook and adjudicated to a final reference label. The SI reports the reliability protocol, aggregate validation metrics, class-wise performance, and a family-level confusion matrix. For the main analyses, I collapse the fine-grained labels into three theoretically motivated families: hard propaganda, soft propaganda, and public service. Hard propaganda includes overtly political and disciplinary content; soft propaganda packages regime-favorable messages through development, culture, tourism, and performance narratives; public service covers instrumental governance content such as procedures, welfare, health, traffic, weather, and emergency guidance. State-governance categories remain in the data for balance checks and appendix material but are not the paper's main theoretical families.

The paper's measurement logic is therefore theory-driven rather than purely unsupervised. The LLM solves the assignment problem at the post level; it does not decide which families matter theoretically. That distinction is important because the theoretical categories come from prior literature on mixed governance communication, soft propaganda, and expressive visibility rather than from an inductive topic model alone.

4 Research Design

The paper exploits two sharp interface discontinuities as natural experiments. On December 21, 2018, WeChat 7.0 replaced the article-bottom "like" button with "good-looking" (*haokan*), bundling low-visibility approval with socially legible recommendation. On July 1, 2020, the "like"

button was restored alongside *zaikan* and a new "share" button, unbundling the two channels. Both changes were platform-wide, simultaneous, and exogenous to individual account or article characteristics.

I estimate sharp regression discontinuity designs centered on each cutoff. The running variable is days relative to the reform date, and the unit of observation is the individual article. All specifications use local linear polynomial estimation with a triangular kernel, MSE-optimal bandwidth selection, and robust bias-corrected confidence intervals following Calonico, Cattaneo, and Titiunik (2014). Standard errors are clustered at the account level (329 prefecture-level accounts) to account for within-account correlation.

Because the running variable is discrete at the day level and the article volume is highly uneven across accounts, the SI re-estimates the main specifications on account-day aggregates so that prolific posting accounts do not dominate the local fit. I also treat the 2017 discovery reform as a contrast case rather than a third main design because it broadened content discovery without directly rebundling the article-footer approval channel; SI SI.16 reports those estimates.

5 Descriptive Findings

5.1 Content, Attention, and Spatial Coverage

Table 2 reports summary statistics for the four content families. Consistent with H1, public service content attracts substantially more routine consumption than any propaganda category: its mean readership is roughly 3.4 times that of hard propaganda and nearly double that of soft propaganda. This ordering is robust across means and medians, confirming that citizens treat official accounts partly as service-delivery interfaces rather than purely as political outlets.

The function of Table 2 is to establish the paper's baseline demand environment before any causal analysis: official accounts are mixed channels in which service content draws systematically higher attention than propaganda content.

Table 3 reports article-level summary statistics for the main engagement variables.

Table 2: Family-level descriptive statistics. Mean and median reads support H1: public service content attracts more routine consumption than propaganda.

Group	Posts (M)	Mean Reads	Median Reads	Likes M (SD)	Shares M (SD)	Zaikan M (SD)
Public Service	1.34	3,550	1,209	230 (732)	14 (56)	137 (469)
Soft Propaganda	1.13	1,889	637	126 (495)	4 (18)	42 (200)
State Governance	0.73	1,295	435	57 (269)	2 (10)	20 (117)
Hard Propaganda	1.00	1,039	344	59 (312)	1 (8)	8 (74)

Table 3: Overall descriptive statistics for article-level engagement variables.

Variable	Min	Max	Mean	Median	SD
Reads	1	100,001	2,113	623	5,162
Likes	0	28,320	131	0	527
Shares	0	2,994	6	0	34
Zaikan	0	17,711	60	0	296

Its function is descriptive calibration. The table makes clear how skewed and heterogeneous these counters are, which in turn motivates the paper's emphasis on engagement rates and local discontinuities rather than raw count comparisons alone.

Figure 1 maps the spatial distribution of corpus coverage at the prefecture level. The dataset covers 329 cities across all 31 provincial-level units. Posting volume concentrates in coastal and densely populated provinces but includes substantial representation from interior and western regions.

The figure's purpose is data provenance rather than hypothesis testing: it shows that the analysis is not driven by a few showcase cities or coastal provinces, but by a broad nationwide archive.

Figure 2 shows that these official accounts remain mixed channels over time rather than converging on a single communicative function. Public service content consistently commands the highest average readership, while propaganda-oriented content remains a persistent share of annual output. The sharp rise in the public-service share around 2020–2022 further underscores that citizens encounter political communication and practical governance information through the same account infrastructure.

The figure's function is to anchor the paper's substantive setting. It shows why engagement outcomes cannot be interpreted against a backdrop of purely political content: the same accounts simultaneously carry service and propaganda content across the full period.

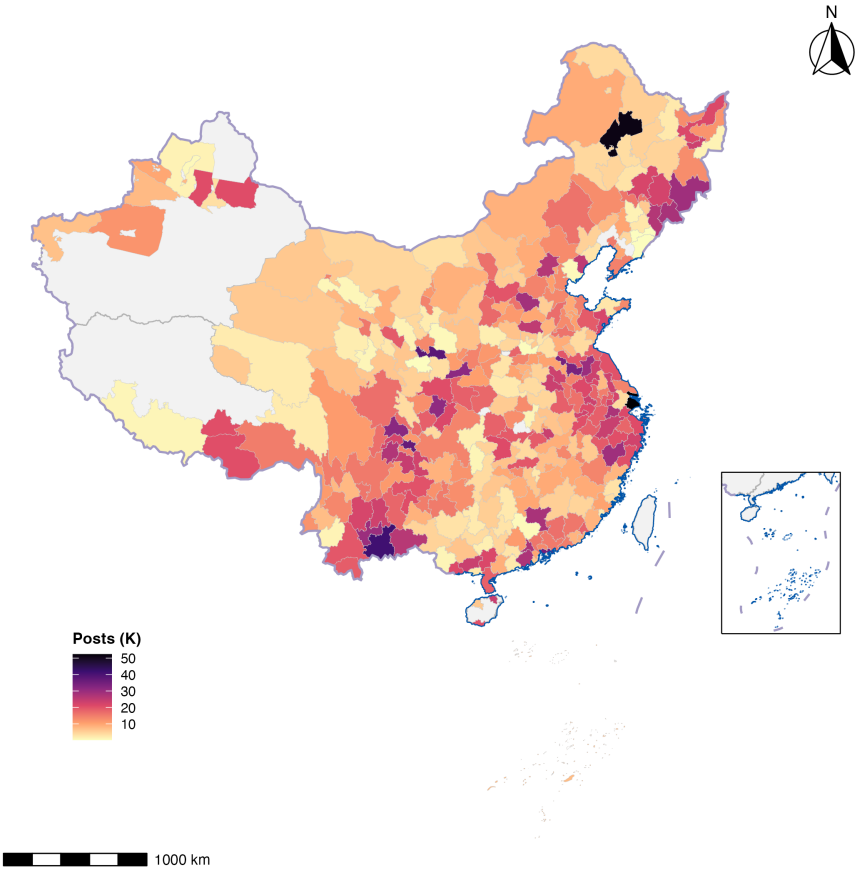

Figure 1: Spatial distribution of total posts across 329 prefecture-level government WeChat accounts.

Note: Shaded prefectures are included in the corpus. Color intensity reflects total post volume (in thousands). Grey prefectures are not covered. Nine-dash line and South China Sea inset included.

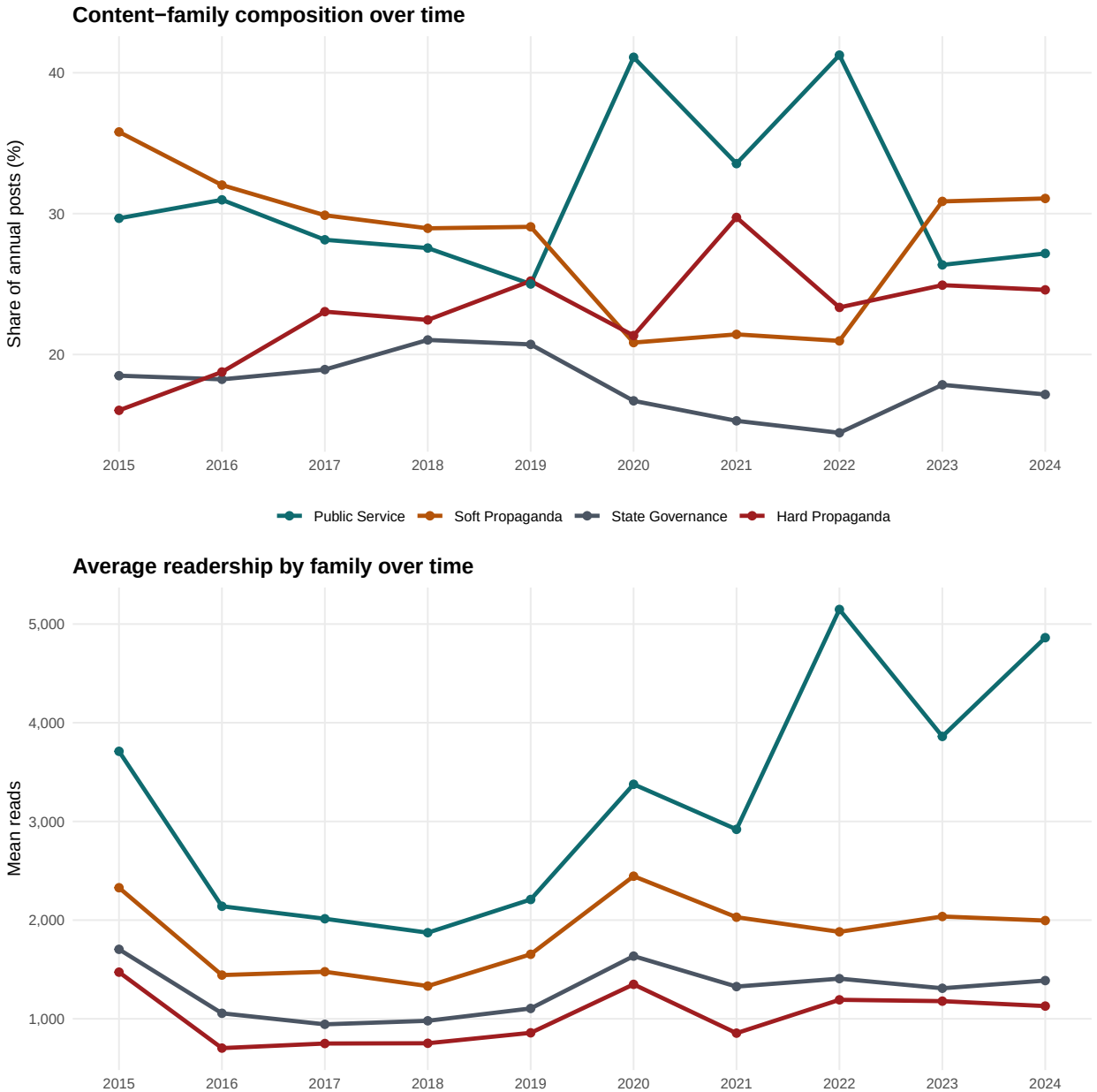

Figure 2: Content mix and readership by family over time.

Note: Top panel reports each family's share of annual posts. Bottom panel reports annual mean readership by family. Public service remains the highest-read family throughout the period, while propaganda-oriented content remains a substantial share of output.

Table 4: Extensive and intensive margins of engagement by interface regime. All values in percent. Extensive: share of posts with any positive count. Intensive: mean rate conditional on positive count.

Regime	Like Ext.	Like Int.	Zaikan Ext.	Zaikan Int.	Share Ext.	Share Int.	N
Pre-reform (before Dec 2018)	41.1	12.6			19.0	1.5	1,022,021
Haokan/Zaikan only (Dec 2018 – Jun 2020)			34.6	11.0	25.2	1.4	667,908
Like + Zaikan + Share (Jul 2020 –)	43.8	12.8	40.2	5.1	27.5	1.3	2,517,143

5.2 Engagement Rates across Interface Regimes

Figure 3 plots monthly mean engagement rates for the three footer-button metrics. Two features stand out. First, the Like rate drops sharply in December 2018 when the button is replaced by Haokan, then recovers in July 2020 when the Like button returns. Second, the Haokan/Zaikan rate appears only after the 2018 redesign and persists after unbundling. Share rates follow a distinct trajectory driven primarily by the addition of the share button to the article footer in 2020.

Its function is diagnostic rather than identificatory: it visually confirms that the timing of the behavioral shifts lines up with the interface reforms before the formal RDD estimates quantify those shifts.

Table 4 decomposes engagement into extensive (any positive count) and intensive (mean rate conditional on participation) margins. The extensive margin for the one-click button drops markedly in the middle regime and recovers after 2020, consistent with the theoretical prediction that raising visibility discourages participation.

This table's function is to clarify the margin of adjustment. It shows that the main pattern is not merely weaker intensity among already-engaged users, but a change in whether users leave any approval trace at all.

6 Causal Findings

6.1 H2: Raising Visibility Reduces One-Click Engagement (2018)

Table 5 reports RDD estimates at the December 2018 cutoff. The main outcome is the *one-click engagement rate*, defined as the sum of like and haokan/zaikan rates, which captures the same

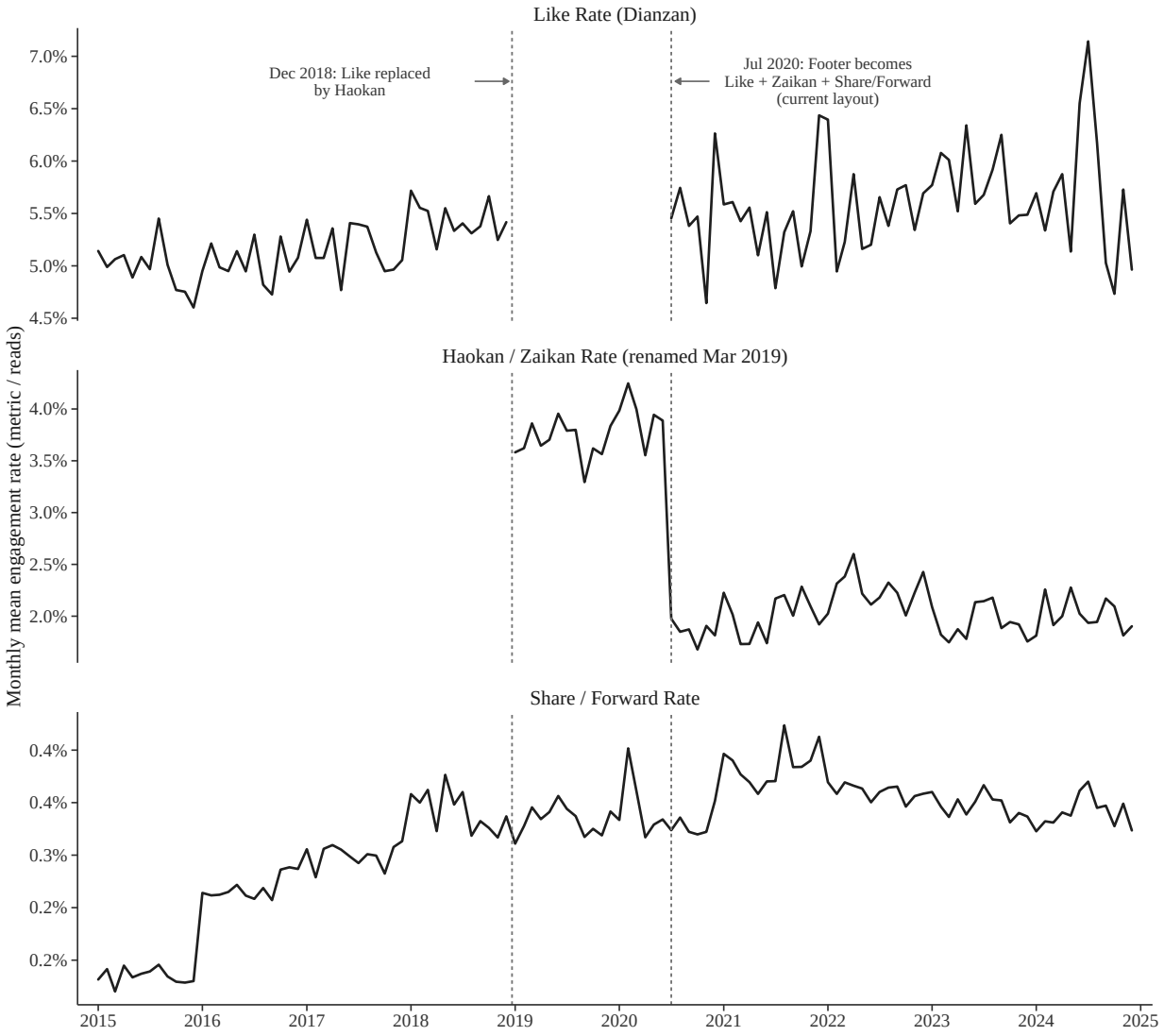

Figure 3: Monthly mean engagement rates by interface regime.

Note: Gaps in the Like panel reflect structural unavailability (button removed Dec 2018 to Jun 2020). Haokan/Zaikan begins Jan 2019. Vertical dashed lines mark the two interface cutoffs. The Haokan/Zaikan panel strip label notes the March 2019 rename.

Table 5: RD estimates at the December 2018 interface reform.

Specification	Estimate	SE	p-value	BW	N (L)	N (R)
One-click (linear, clustered)	-0.0193	0.0038	3.641e-07	8	7,972	9,226
One-click (quadratic, clustered)	-0.0182	0.0056	0.001	9	9,110	9,794
One-click (robust HC, clustered)	-0.0193	0.0038	3.620e-07	8	7,972	9,226
Share rate (placebo, clustered)	4.1449e-05	0.0006	0.945	10	9,110	9,794

Table 6: RD estimates at the July 2020 interface reform.

Specification	Estimate	SE	p-value	BW	N (L)	N (R)
One-click rate (linear, clustered)	0.0289	0.0039	6.465e-14	4	5,561	6,155
Like rate (linear, clustered)	0.0431	0.0018	3.312e-124	7	7,375	9,211
Zaikan rate (linear, clustered)	-0.0118	0.0023	2.554e-07	6	7,375	9,211
Share rate (linear, clustered)	0.0001	0.0004	0.791	12	15,866	17,151

footer-button slot across regimes.

The function of this table is formal identification: it provides the paper's main estimate of what happens when low-visibility approval is rebundled into a more public signal.

The preferred linear specification with account-clustered standard errors yields an estimate of -0.019 ($p < 0.001$): one-click engagement drops by roughly 1.9 percentage points when the button becomes socially visible. The effect is robust to quadratic polynomial order and heteroskedasticity-consistent standard errors. Importantly, the share rate shows no discontinuity at this cutoff ($p = 0.945$), confirming that the effect is specific to the one-click footer button.

Figure 4 serves a complementary function to Table 5: it lets the reader inspect the local discontinuity directly and see that the identifying variation is concentrated at the cutoff rather than driven by a broader trend break.

6.2 H3: Unbundling Restores Low-Visibility Approval (2020)

Table 6 reports estimates at the July 2020 cutoff, where the platform unbundled the single haokan/zaikan button into separate "like" and "zaikan" buttons.

Its function is to separate recovery in total approval from reallocation across channels. Because the 2020 reform changes the menu of visible and less-visible actions, the table reports both the aggregate and the components.

The total one-click rate increases by 2.9 percentage points ($p < 0.001$). The restored like channel alone rises by 4.3 percentage points ($p < 0.001$), while zaikan falls by 1.2 percentage

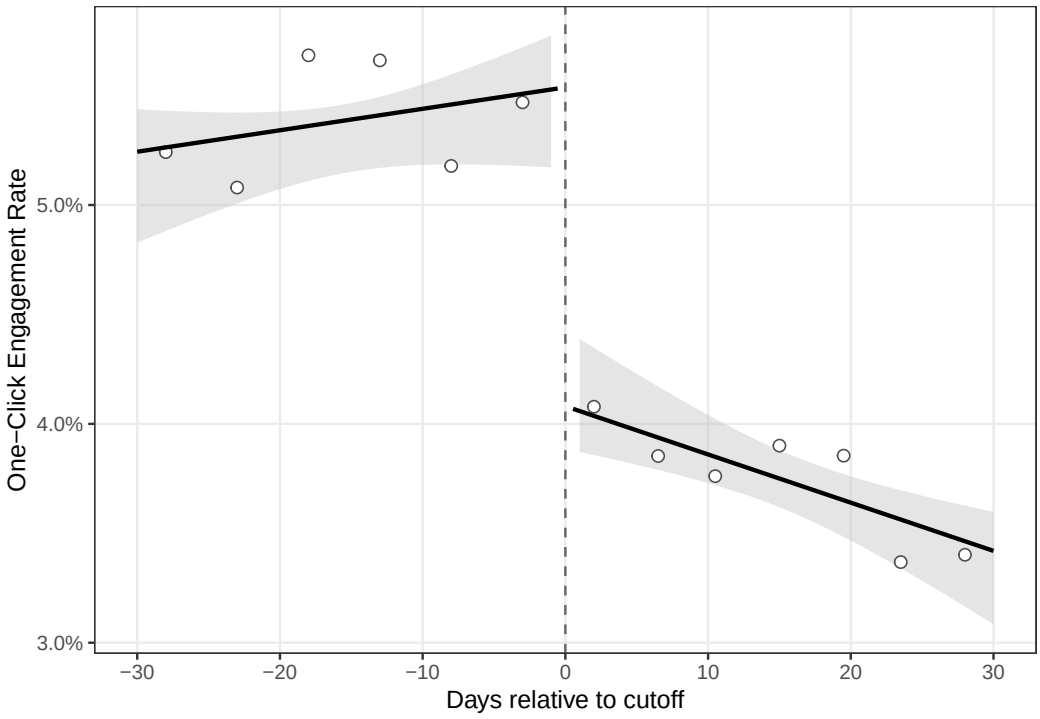

Figure 4: RDD plot at December 2018: one-click engagement rate.

Note: Binned daily means (open circles), local linear polynomial fit (solid lines), and 95% confidence intervals (grey bands) from daily-aggregated data. Vertical dashed line marks the cutoff. One-click rate = like rate (pre-reform) + haokan rate (post-reform).

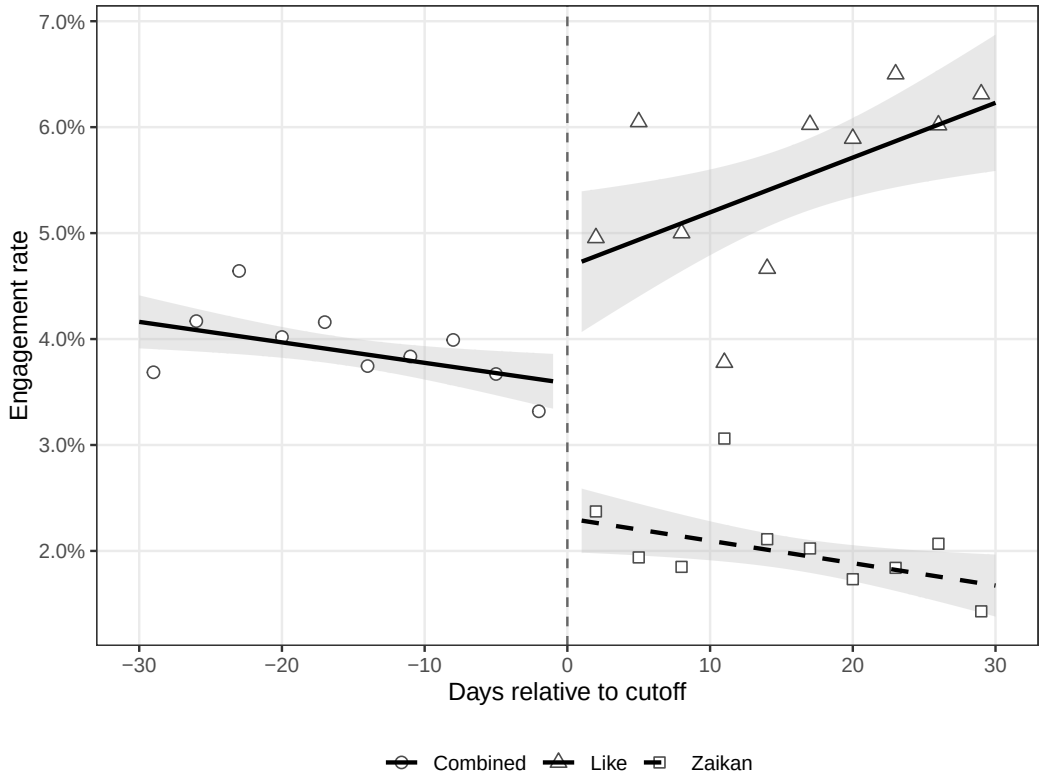

Figure 5: RDD at July 2020: unbundling one-click engagement into like and zaikan.
Note: Left of cutoff: combined one-click rate (circles, solid line). Right of cutoff: like rate (triangles, solid line) and zaikan rate (squares, dashed line). Grey bands show 95% CIs from daily-aggregated weighted linear fits.

points ($p < 0.001$) as some approval shifts away from the high-visibility channel. The share rate is unaffected ($p = 0.791$).

Figure 5 visualizes the unbundling directly. Before the cutoff, only the combined one-click rate is observed (zaikan alone). After the cutoff, engagement splits into a dominant like component and a smaller but persistent zaikan component.

The figure's function is decomposition: instead of merely reporting that total one-click engagement rises, it shows how that rise is produced by the restoration of a lower-visibility like channel and a simultaneous contraction in the more public zaikan channel.

Table 7: Heterogeneous RD estimates by content family at the July 2020 cutoff.

Specification	Estimate	SE	p-value	BW	N (L)	N (R)
Public Service: One-click	0.0378	0.0059	1.698e-10	5	2,586	3,045
Public Service: Like	0.0442	0.0032	2.085e-44	6	2,956	3,733
Public Service: Zaikan	-0.0046	0.0044	0.306	5	2,586	3,045
Soft Propaganda: One-click	0.0311	0.0081	1.173e-04	6	1,393	1,537
Soft Propaganda: Like	0.0434	0.0037	3.354e-32	6	1,659	1,794
Soft Propaganda: Zaikan	-0.0139	0.0038	2.793e-04	12	3,587	3,601
State Governance: One-click	0.0259	0.0054	1.304e-06	5	1,108	1,477
State Governance: Like	0.0338	0.0031	5.170e-27	6	1,108	1,477
State Governance: Zaikan	-0.0094	0.0034	0.006	7	1,217	1,718
Hard Propaganda: One-click	0.0245	0.0064	1.193e-04	5	1,284	1,644
Hard Propaganda: Like	0.0472	0.0032	4.709e-50	6	1,543	1,966
Hard Propaganda: Zaikan	-0.0174	0.0031	2.334e-08	9	2,316	2,563

6.3 H4: Political-Risk Gradient

Table 7 reports heterogeneous RDD estimates at the 2020 cutoff by content family. The theoretical prediction is that reallocation from high-visibility zaikan to low-visibility like should be strongest for politically risky content.

Its function is to connect the interface argument back to political content. If visibility costs matter politically rather than mechanically, the size of the shift should vary systematically with content risk.

The results confirm the gradient. The zaikan drop is largest for hard propaganda (-0.017, $p < 0.001$), followed by soft propaganda (-0.014, $p < 0.001$) and state governance (-0.009, $p = 0.006$). For public service content, the zaikan drop is small and not statistically significant (-0.005, $p = 0.306$). This ordering is consistent with the theory: citizens are most reluctant to leave socially visible endorsement traces on politically charged content, and most willing to shift approval to the restored low-visibility channel when the expressive risk is highest.

Figure 6 visualizes the 2020 unbundling separately by content family. The visible zaikan component remains especially compressed for hard and soft propaganda, while public service retains a larger socially legible component. That pattern sharpens the interpretation of H4: what varies across families is not simply total engagement, but the willingness to leave behind a visible endorsement trace.

The figure's function is interpretive. It shows why the family-specific table matters substantively: the core difference across families lies in the composition of approval between visible and

less-visible traces, not just in overall clicking.

6.4 Mechanism: Behavioral Self-Censorship and Visibility Avoidance

The 2020 redesign makes it possible to quantify the mechanism more directly. The main 2020 table uses outcome-specific MSE-optimal bandwidths, which is appropriate for separate estimation but not for an exact decomposition. To recover an additive decomposition, I re-estimate the July 2020 cutoff with a *shared* local-linear bandwidth inherited from the total one-click specification. Let $\hat{\mu}_m^-(h)$ and $\hat{\mu}_m^+(h)$ denote the left- and right-limit fitted values at the cutoff for outcome $m \in \{O, L, Z\}$, where O is total one-click engagement, L is like, and Z is *zaikan*. Using the common bandwidth h, I proceed in three steps: estimate the same local-linear specification for all three outcomes, define the shared-bandwidth jump as

$$\hat{\tau}_m(h) = \hat{\mu}_m^+(h) - \hat{\mu}_m^-(h),$$

and then convert those three jumps into descriptive mechanism summaries. Because $O = L + Z$ at the article level and all three outcomes are estimated with the same bandwidth, kernel, and local polynomial order, the decomposition satisfies

$$\hat{\tau}_O(h) = \hat{\tau}_L(h) + \hat{\tau}_Z(h).$$

I summarize the decomposition with two quantities. First, I define a *behavioral self-censorship proxy* as the share of the post-2020 increase in low-visibility likes that shows up as net-new one-click expression rather than mere substitution away from visible endorsement:

$$\text{Behavioral self-censorship proxy} = \frac{\hat{\tau}_O(h)}{\hat{\tau}_L(h)}, \qquad \text{Visible-to-private substitution share} = \frac{-\hat{\tau}_Z(h)}{\hat{\tau}_L(h)}.$$

Second, for each content family f, I define a *visibility-avoidance index* (VAI) as

$$\widehat{VAI}_f = \frac{\hat{\mu}_{L,f}^+(h)}{\hat{\mu}_{L,f}^+(h) + \hat{\mu}_{Z,f}^+(h)}.$$

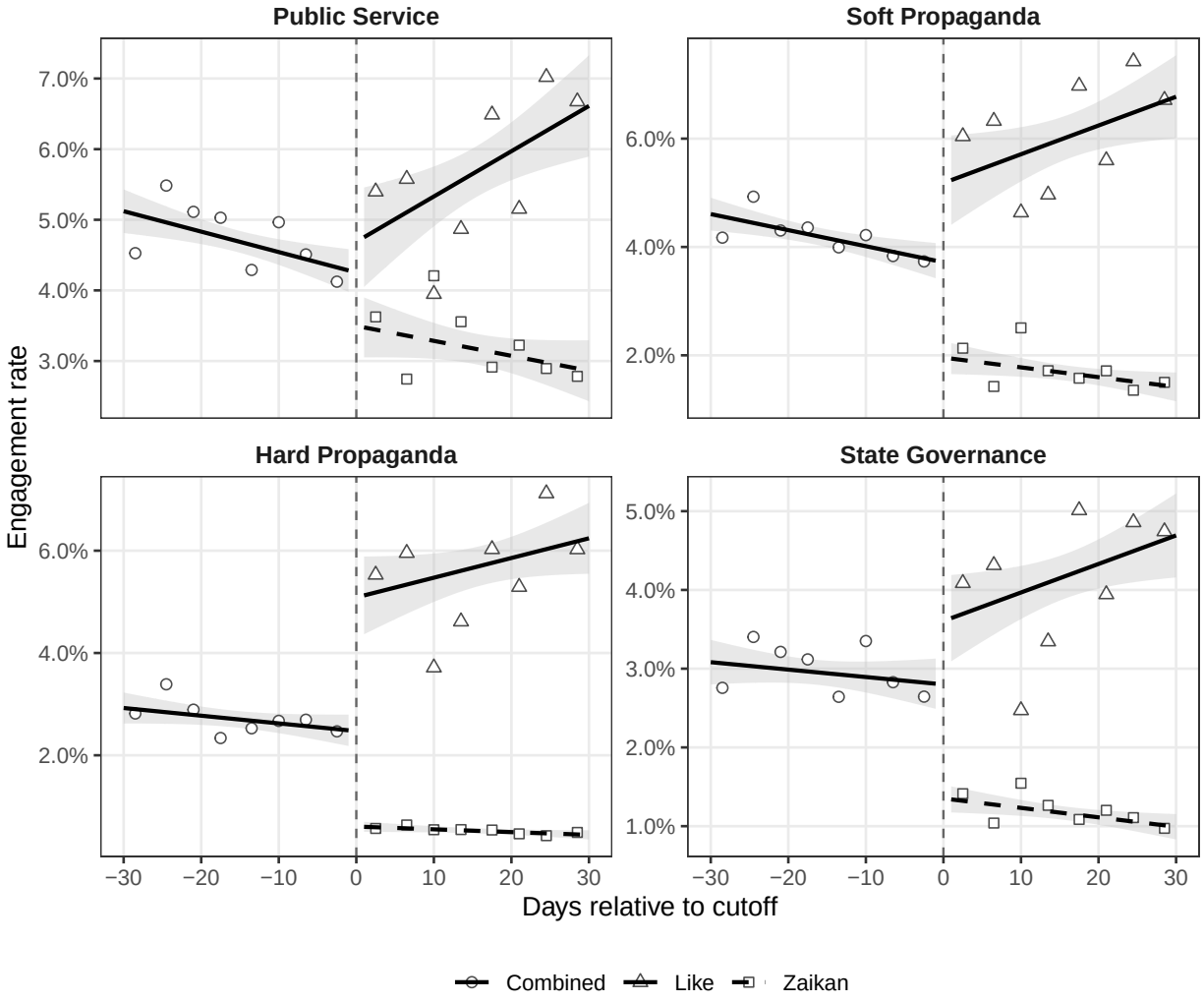

Figure 6: Unbundling of one-click engagement at the July 2020 cutoff, by content family.

Note: In each panel, circles to the left of the cutoff show the pre-reform combined one-click rate; triangles and squares to the right show the post-reform like and zaikan components. Lines are local linear fits with 95% confidence bands from daily-aggregated data.

Table 8: Shared-bandwidth decomposition of the July 2020 redesign. All three component RDDs are re-estimated with a common local-linear bandwidth inherited from the one-click specification (h = 4.35 days), triangular weights, and account-clustered robust standard errors. Derived ratios are descriptive summaries rather than separate causal estimands.

Statistic	Estimate	SE	p-value
Common bandwidth h (days)	4.35		
Shared-bandwidth one-click jump (tau_O, pp)	3.05	0.27	1.552e-29
Shared-bandwidth like jump (tau_L, pp)	4.29	0.18	1.851e-123
Shared-bandwidth zaikan jump (tau_Z, pp)	-1.24	0.21	4.905e-09
Right-cutoff like mean (mu_L+, pp)	4.29		
Right-cutoff zaikan mean (mu_Z+, pp)	1.99		
Behavioral self-censorship proxy (tau_O / tau_L, %)	71.04		
Visible-to-private substitution share (-tau_Z / tau_L, %)	28.96		
Post-cutoff visibility-avoidance index (VAI, %)	68.33		

This is the share of post-reform one-click approval routed through the lower-visibility like channel rather than publicly legible *zaikan*. Higher values indicate stronger avoidance of visible endorsement conditional on leaving any one-click trace. I treat these quantities as descriptive mechanism summaries rather than as a true preference-falsification rate.

Table 8 reports the aggregate decomposition. With the shared bandwidth, the one-click jump is 3.05 percentage points, the like jump is 4.29 points, and the *zaikan* jump is -1.24 points, so the additive identity holds exactly. The behavioral self-censorship proxy is 71.0%, meaning that about seven-tenths of the restored low-visibility approval appears as net-new one-click expression rather than simple migration out of visible *zaikan*. The remaining 29.0% is visible-to-private substitution. The post-cutoff VAI is 68.3%, indicating that more than two-thirds of one-click approval at the cutoff is routed through the less visible channel.

Its function is mechanism quantification. Instead of stopping at the statement that likes rise and *zaikan* falls, the table translates the 2020 redesign into a lower-claim but directly interpretable behavioral proxy for self-censorship and visible-trace avoidance.

Figure 7 and Table 9 extend the VAI to the four content families. Public service has the lowest VAI at 52.7%, meaning approval remains relatively balanced between low- and high-visibility channels. Hard propaganda has by far the highest VAI at 88.2%, implying that nearly nine in ten one-click approval traces are routed through likes rather than *zaikan*. State governance and soft propaganda lie in between, both well above public service. This pattern is exactly what the visibility-cost argument predicts: once content becomes politically riskier, users who still engage

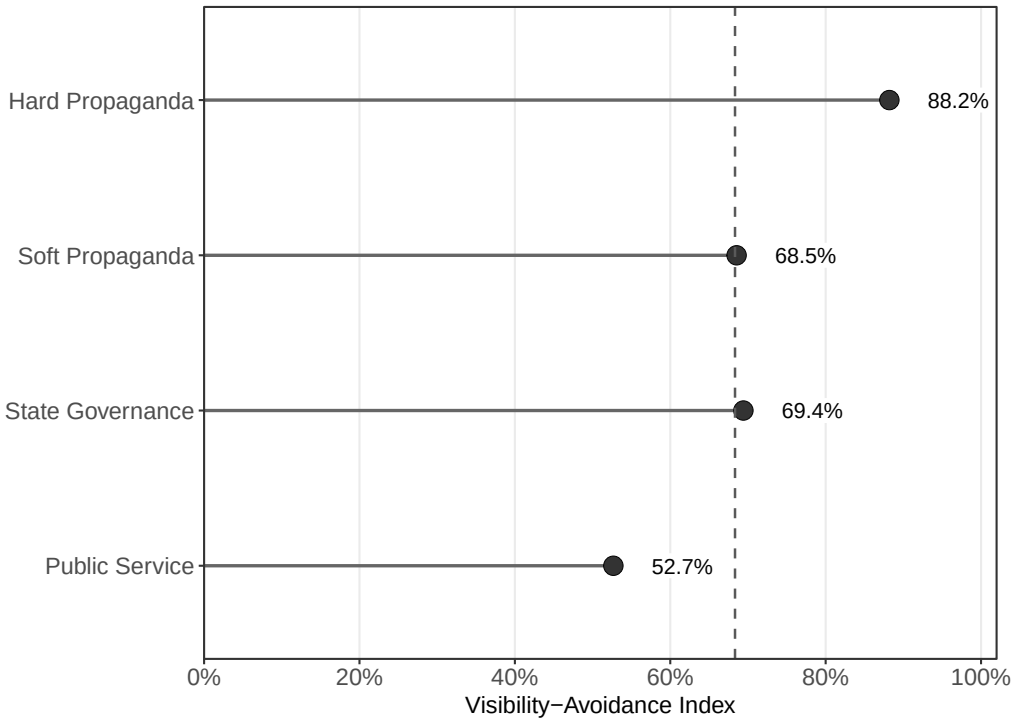

Figure 7: Visibility-avoidance index by content family at the July 2020 cutoff.

Note: The visibility-avoidance index is defined as $\widehat{VAI}_f = \hat{\mu}^+_{L,f}(h)/[\hat{\mu}^+_{L,f}(h) + \hat{\mu}^+_{Z,f}(h)]$, where the right-side fitted values are estimated with the common local-linear bandwidth from the total 2020 one-click specification. The dashed vertical line marks the overall post-cutoff VAI.

disproportionately choose the less visible channel.

Their function is to make the risk gradient legible in a single mechanism metric. The family-specific decomposition table shows the ingredients of the index, while the figure makes clear that the main contrast is between public-service content and politically riskier material, with hard propaganda as the extreme case of visible-endorsement avoidance.

Table 9: Visibility-avoidance index by content family at the July 2020 cutoff. For each family, the post-cutoff like and zaikan means are the right-side local-linear fitted values at x = 0 using the common bandwidth h = 4.35 days. VAI = like / (like + zaikan).

Family	Post-cutoff like (pp)	Post-cutoff zaikan (pp)	VAI (%)
Public Service	4.37	3.92	52.7
State Governance	3.31	1.46	69.4
Soft Propaganda	4.35	2.00	68.5
Hard Propaganda	4.78	0.64	88.2

Table 10: Placebo outcomes at each cutoff. Reads and collect rate should be unaffected by footer-button reforms.

Specification	Estimate	SE	p-value	BW	N (L)	N (R)
Reads (2018)	245.1400	207.7260	0.238	9	9,110	9,794
Reads (2020)	212.6953	270.2765	0.431	8	10,831	12,380
Collect Rate (2018)	0.0009	0.0006	0.154	7	6,838	8,103
Collect Rate (2020)	8.2736e-05	0.0003	0.808	11	13,281	14,861
Share Rate (2018)	4.1449e-05	0.0006	0.945	10	9,110	9,794
Share Rate (2020)	0.0001	0.0004	0.791	12	15,866	17,151

6.5 Mechanism Check: Visibility, Not Attention

The core theoretical claim is that interface reforms changed the *visibility cost* of engagement, not the underlying attention to official content. If the mechanism operates through visibility rather than attention, predetermined outcomes that do not depend on footer-button design should show no discontinuity.

Table 10 reports placebo tests at both cutoffs. Reads, collect rate (accessed via the top-right menu, not the footer), and share rate at the 2018 cutoff all show no significant jump. This confirms that the reform affected only the footer-button engagement channel, consistent with a pure visibility mechanism.

The table's function is mechanism discrimination. It helps rule out a generic attention shock by showing continuity in outcomes that should not respond immediately to footer-button redesign.

6.6 Validity

Figure 8 presents McCrary density tests at both cutoffs. The distribution of daily post counts shows no discontinuity, ruling out strategic manipulation of posting timing around the reform dates.

The function of this figure is identification validity. It addresses the concern that accounts might have sorted publication timing around the reform dates in ways that would mechanically create or amplify discontinuities.

Table 11 reports covariate balance tests. Mean reads, hard propaganda share, and public service share are all continuous at both cutoffs, confirming that content composition did not shift at the reform dates.

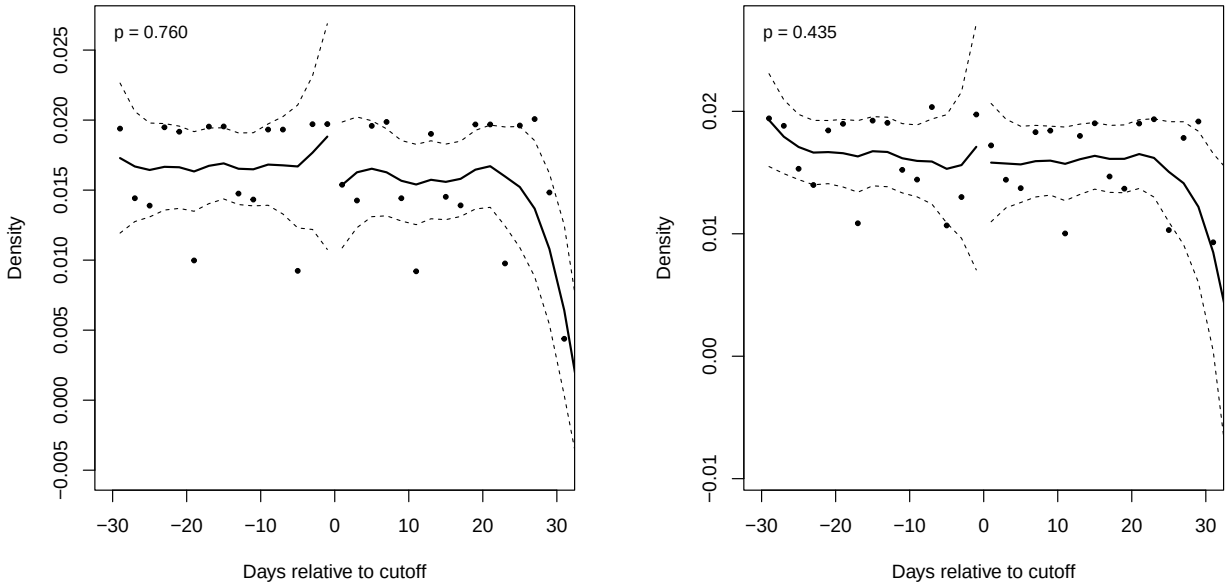

Figure 8: McCrary density tests at each cutoff.

Note: Binned daily post-count densities within ±30 days of each cutoff, with local polynomial density estimates and 95% confidence bands fitted separately on each side. Reported p-values from the Cattaneo-Jansson-Ma density continuity test appear in the upper-left corner of each panel.

Table 11: Covariate balance at each cutoff. Account-clustered SEs.

Specification	Estimate	SE	p-value	BW	N (L)	N (R)
Reads (2018)	245.1400	207.7260	0.238	9	9,110	9,794
Reads (2020)	212.6953	270.2765	0.431	8	10,831	12,380
Hard prop. (2018)	-0.0387	0.0206	0.060	9	9,110	9,794
Hard prop. (2020)	0.0113	0.0221	0.609	7	9,164	10,811
Pub. service (2018)	-0.0234	0.0247	0.342	5	4,623	4,624
Pub. service (2020)	0.0152	0.0234	0.517	6	7,375	9,211

The function of the balance table is narrower than that of the McCrary test. Rather than asking whether posting density itself changes, it asks whether the local mix of content and baseline attention shifts at the cutoff in ways that would complicate interpretation.

The SI extends these validity checks in three directions. SI SI.19 reports richer composition diagnostics, including title length, head-article share, and additional content shares. SI SI.17 shows that the main results survive account-day aggregation, and SI SI.18 shows that the 2020 estimates are similar after excluding COVID- and emergency-linked posts or excluding Hubei and Wuhan accounts.

7 Conclusion

Official engagement metrics on authoritarian platforms are not transparent measures of support. They are shaped by mixed content environments and by the visibility structure of the interface itself. On WeChat, the central analytical task is therefore to separate attention from endorsement and to study how interface design conditions the forms of engagement that users are willing to leave behind.

The broader implication is that digital authoritarian control works partly by structuring traces, not only by changing beliefs. Once we distinguish low-risk consumption from low-visibility approval and from socially legible endorsement, the political meaning of a click changes. That insight extends preference falsification theory to platform-mediated behavior and suggests a more cautious reading of engagement metrics in authoritarian settings.

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Online Supplemental Information

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A WeChat, Public Accounts, and Interface Reforms

A.1 Platform Scale and Official-Account Infrastructure

Weixin and its international sister product WeChat are not merely messaging applications. They are a broad digital infrastructure that combines chat, social networking, public content distribution, payments, mini programs, and search within the same app environment. Tencent's 2024 annual results report 1.385 billion combined monthly active accounts as of 31 December 2024, while Tencent's product materials describe the platform as an ecosystem spanning Moments, Official Accounts, Weixin Pay, Mini Programs, Search, Channels, and related services (Tencent 2026b; Tencent Holdings Limited 2025). For this paper, the most important layer is the official-account system. Tencent's own description of Weixin Official Accounts presents it as a publishing and notification infrastructure through which individuals and organizations gather followers, push content, and send service or marketing notifications. The same page reports more than 20 million official accounts as of February 2021 and distinguishes between subscription accounts, oriented toward content publication and readership management, and service accounts, oriented toward user management and customer service (Tencent 2026a).

A.2 Why Public Accounts Matter for Local Government

This architecture matters for local-government communication because official accounts are embedded inside ordinary social and administrative life rather than separated into a standalone news app. A user may encounter a government article through a subscription feed, a chat share, a Moments post, a search result, or a recommendation channel such as *kanyikan* ("See"). Once opened, the article sits inside the same environment that also contains private chats, group chats, payments, and service tools. That is why official accounts can function at once as broadcast channels, low-friction service interfaces, and distribution nodes inside the larger WeChat ecosystem. Tencent's 2017 first-quarter results, for example, already emphasized that official-account content was being tied more closely to search and mini-program traffic through embedded links and discovery tools (Tencent Holdings Limited 2017). Existing research on government WeChat accounts reaches a similar conclusion from the user side: many citizens treat these accounts as routine lookup and service tools rather than as purely political venues (Jiang et al. 2021; Liu, Ma, and Xia 2024; Yu and Shen 2026). For the purposes of this paper, the key implication is that article-level interaction takes place inside a platform where private communication, semi-public recommendation, and public-facing service use are tightly bundled together.

A.3 Interface Reforms Relevant to Engagement Measures

The article-level engagement measures used in this paper were shaped by a small number of platform redesigns that changed what users could do from the bottom of a public-account article page. First, contemporary reporting from late July 2014 documented that WeChat began displaying article-level read counts and a visible like button on public-account pages, creating the first widely visible article-side counters (People.cn 2014). Second, in May 2017, WeChat introduced "Search" and "See" through WeChat Lab and the discovery tab, broadening how users could encounter official-account content beyond simple subscription order (Sina Tech 2017; Tencent Hold-

Table SI.1: Year-level descriptive summary of posting volume and engagement.

year	Posts (M)	Mean Reads	Median Reads	One-Click (%)	Like/Read (%)	Zaikan/Read (%)	Share/Read (%)	10w+ (%)
2015	0.17	2,486	883	4.97	4.97	0.00	0.18	0.006
2016	0.24	1,450	468	5.02	5.02	0.00	0.27	0.025
2017	0.28	1,359	438	5.16	5.16	0.00	0.30	0.027
2018	0.34	1,277	413	5.39	5.27	0.12	0.34	0.028
2019	0.36	1,478	486	3.69	0.00	3.69	0.33	0.025
2020	0.54	2,459	815	5.48	2.45	3.02	0.34	0.060
2021	0.54	1,871	566	7.44	5.44	2.01	0.39	0.050
2022	0.61	2,999	864	7.80	5.54	2.26	0.36	0.111
2023	0.58	2,174	647	7.73	5.80	1.94	0.35	0.036
2024	0.56	2,457	674	7.69	5.68	2.01	0.34	0.057

ings Limited 2017). Third, on 21 December 2018, WeChat 7.0 replaced the older like button with *haokan* ("good-looking"), and press coverage at the time emphasized that clicking *haokan* could sync the article into the "See" feed and make the signal visible to friends (Zhoudao 2018). Fourth, on 15 March 2019, *haokan* was renamed *zaikan* (roughly, "seen" or "reading now"); coverage framed the shift as softer wording for the same socially visible recommendation channel rather than as a new interaction type (TechNode 2019). Finally, on 29 June 2020, the article footer was redesigned again: like returned alongside *zaikan*, and a dedicated share button was added, making it easier to send articles to friends or Moments while also restoring a lower-visibility approval channel (Securities Times 2020).

Taken together, these reforms changed not only how official-account content circulated but also the social meaning of leaving a trace on an article page. In this paper's design, 2017 therefore serves mainly as a discovery and distribution reform, 2018 as a visibility-regime change in one-click approval, 2019 as a useful rename placebo, and late June 2020 as the return of a distinct lower-visibility like channel.

A.4 Sample Construction and Coverage

The working archive contains the universe of posts published by 329 prefecture-level government public accounts between 1 January 2015 and 31 December 2024, for a total of 4,207,072 posts. The preserved nationwide fields are account identifier, publication date, title, and article-side engagement counters. Because article bodies are not consistently retained in the working archive, title text is the stable unit for full-corpus measurement and classification. The validation audit then draws a separate 2,000-title gold-standard sample for dual human coding and adjudication.

Table SI.1 reports the year-level composition of the corpus and the main engagement measures. Posting volume expands steadily after 2018, especially during 2020–2022, while the share of top-coded "100k+" articles remains small relative to the full corpus.

This table's function is scope documentation. It tells the reader how the archive evolves over time and shows that the paper's outcomes are observed across a large and expanding corpus rather than a thin event-study slice.

Table SI.2 summarizes the spatial coverage of the archive. The corpus spans all 31 provincial-level units and 329 prefecture-level cities, with broad variation in posting volume across cities.

Its function is complementary to the year table: it documents the breadth of spatial coverage and the degree of cross-city variation that later heterogeneity and robustness checks draw on.

Figure SI.1 complements the post-count map in the main text by showing average readership

Table SI.2: Spatial coverage and posting concentration across provinces and prefecture-level cities.

Statistic	Value
Province-level units covered	31
Prefecture-level cities covered	329
Posts per city: mean	12,672
Posts per city: median	11,228
Posts per city: sd	8,899
Posts per city: min	169
Posts per city: max	52,380

across covered cities. Consumption is uneven across space, but the archive includes substantial variation both within and across provinces.

The figure's function is descriptive context. It makes clear that the corpus varies not only in how much governments post, but also in how much attention their audiences appear to pay.

A.5 Engagement Counters and Measurement Choices

The article footer does not present a fixed menu of counters throughout the sample period. Before December 2018, the footer displayed a like button. From late December 2018 through June 2020, that slot became *haokan*/*zaikan*, which carried a more socially legible recommendation meaning because it linked to the "See" feed. From late June 2020 onward, the footer displayed both like and *zaikan*, and also added a dedicated share button. Collect remained accessible through the top-right menu rather than the footer, which is why it serves as a useful placebo outcome in the main text.

These interface shifts motivate the paper's outcome construction. The main 2018 and 2020 outcome is therefore the *one-click engagement rate*, defined as like plus *haokan*/*zaikan* whenever those channels occupy the same approval slot across regimes. That aggregation allows the design to separate changes in total approval from changes in how approval is allocated between lower-visibility and higher-visibility channels after 2020. Read counts require one additional harmonization step because WeChat top-codes some counters at "100k+"; throughout the paper I recode those observations to 100,001 so they remain ordered while preserving the censoring boundary.

B WeChat Title Data and Classification Validation

This section documents the title-based WeChat classification pipeline used in the main analyses. The cleaned corpus covers all posts published by 329 prefecture-level government WeChat public accounts between 1 January 2015 and 31 December 2024. The key measurement constraint is that the archived source file preserves post titles consistently across the full nationwide collection but does not preserve article bodies in the current working archive. The unit of text classification is therefore the individual post title. In addition to the title itself, the working archive preserves auxiliary text-side cues such as a short keyword field, a short reason field, and a confidence score. These metadata are used only as classification inputs. Engagement outcomes such as reads, likes, shares, and *zaikan* never enter the classification prompt.

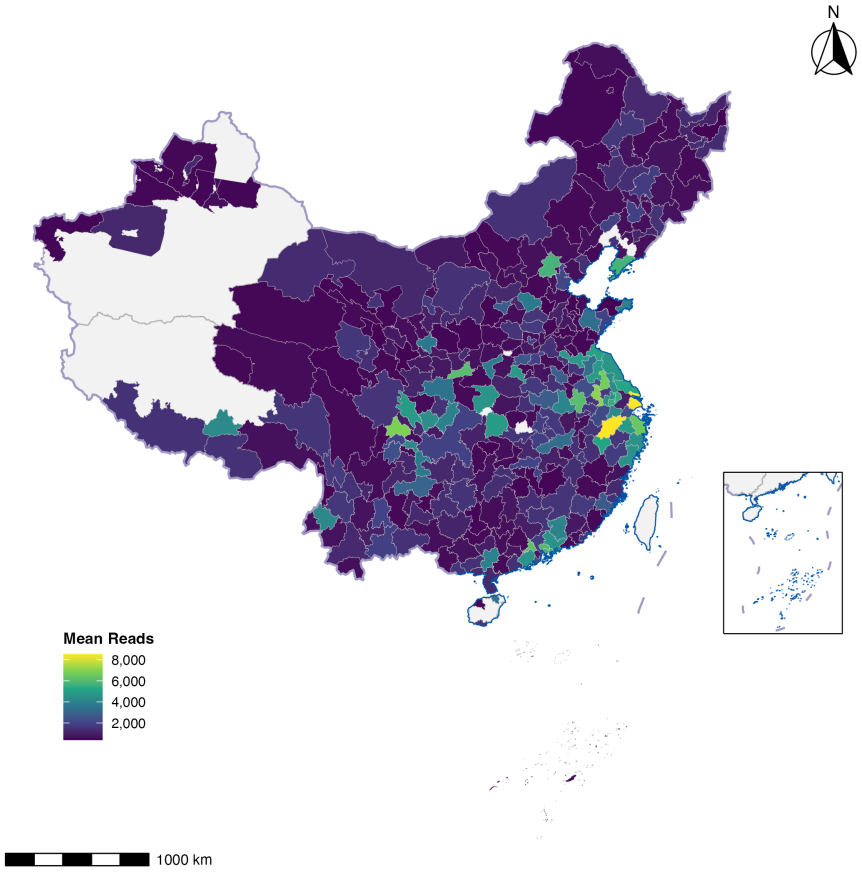


Figure SI.1: Spatial distribution of mean article readership across covered cities.

Note: Shaded prefectures are included in the corpus. Color intensity reflects mean article reads rather than total posting volume. Grey prefectures are not covered.

Table SI.3: Final ten-label taxonomy, higher-order families, and corpus shares.

detailed_label	family	n_posts	share_posts
Ideological Education	Hard Propaganda	470,939	0.112
Leadership and Political Events	Hard Propaganda	530,382	0.126
Public Service Information	Public Service	524,185	0.125
Social Protection and Welfare	Public Service	274,601	0.065
Emergency and Risk Communication	Public Service	544,084	0.129
Policy and Administrative Disclosure	State Governance	432,057	0.103
Social Governance and Enforcement	State Governance	203,340	0.048
Mass Mobilization and Participation	State Governance	96,840	0.023
Economic Development and Construction	Soft Propaganda	641,160	0.152
City Image and Cultural Activity	Soft Propaganda	489,484	0.116

Table SI.4: Discovery-sample design used to refine category boundaries and codebook rules.

phase	target	rule
Round 1	200	Balanced grid across five time bins and the ten cleaned labels
Round 2A	50	Short or low-information titles
Round 2B	50	Low-confidence titles
Round 2C	50	Mixed-domain titles spanning multiple coding cues
Round 2D	50	Adjacent-boundary titles from semantically neighboring categories

B.1 Taxonomy Construction and Discovery Review

The final taxonomy fixes ten mutually exclusive detailed labels and four higher-order families. Table SI.3 reports the final taxonomy, the number of posts assigned to each label, and each label's share of the cleaned corpus.

The codebook was finalized after a 400-title discovery review carried out with experienced media practitioners to surface both common labels and difficult boundary cases. As Table SI.4 shows, the first round balanced across five time bins and the ten cleaned labels, while the second round concentrated on short or low-information titles, low-confidence titles, mixed-domain titles, and adjacent boundary cases.

That discovery review clarified the main inclusion and exclusion rules that later structure both human coding and model prompting. The hardest boundaries are not random. They recur where title wording mixes leadership reporting with development claims, routine service information with welfare substance, policy release with enforcement, emergency warning with ordinary governance control, and development performance with city branding. Table SI.5 summarizes the resulting coding cues and the corresponding title patterns.

That discovery review clarified the boundary rules later used in both human coding and model prompting. Its role was to stabilize the codebook and the main-communicative-purpose rule before full-corpus assignment rather than to supply any supervised training set in the machine-learning sense.

B.2 Human Coding Rules and Validation Sample

The validation audit drew 2,000 titles with sampling probabilities proportional to city-level corpus size, while retaining at least one title from every city in the corpus. This preserves exposure to the largest posting environments without collapsing the validation sample into only a handful of high-volume cities. Table SI.6 summarizes the resulting audit sample.

Two independent coders then assigned each title one and only one primary label and the corresponding higher-order family. The operative rule was to code the title's *main communicative*

Table SI.5: Illustrative coding cues and title patterns for the ten detailed labels.

detailed_label	family	decision_cue	example_title_pattern
Ideological Education	Hard Propaganda	Political learning, Party education, discipline, and value transmission	"Study and implement the Party Congress spirit" style political-study or Party-education title
Leadership and Political Events	Hard Propaganda	Leaders, meetings, inspections, speeches, appointments, and deployments	"Municipal Party secretary inspects major projects" style leader-activity or meeting report
Public Service Information	Public Service	Practical guidance, reminders, procedures, notices, and daily-use service information	"Housing provident fund application guide" style practical procedure or reminder title
Social Protection and Welfare	Public Service	Education, health, employment, subsidies, pensions, and welfare provision	"Medical reimbursement policy adjustment" style welfare, education, health, or subsidy title
Emergency and Risk Communication	Public Service	Warnings, hazards, disasters, epidemics, safety alerts, and urgent risk communication	"Blue rainstorm warning" style safety warning, hazard, or epidemic-risk title
Policy and Administrative Disclosure	State Governance	Policy texts, formal notices, administrative disclosure, budgets, and statistics	"2024 government work report" style policy-release or administrative-disclosure title
Social Governance and Enforcement	State Governance	Enforcement, punishment, inspections, anti-fraud, and order-maintenance action	"Special rectification campaign bulletin" style enforcement or rectification title
Mass Mobilization and Participation	State Governance	Calls for participation, volunteering, campaigns, voting, and civic mobilization	"Civilized city creation initiative" style participation or volunteering title
Economic Development and Construction	Soft Propaganda	Projects, investment, industry, infrastructure, agriculture, and development performance	"Major projects break ground" style development-performance or infrastructure title
City Image and Cultural Activity	Soft Propaganda	Tourism, culture, festivals, exhibitions, place branding, and local storytelling	"Cherry blossom festival opens" style culture, tourism, festival, or branding title

Table SI.6: Coverage summary for the 2,000-title validation sample.

metric	value
Annotated titles	2,000
Cities represented	329
Provincial-level units represented	31
Minimum titles per city	1
Median titles per city	5
Maximum titles per city	25

purpose, not the broadest noun phrase or the most politically salient word in isolation. In practice, this meant preferring public-service information over welfare only when the title mainly explained how to do something; preferring leadership and political events over policy disclosure when leaders, meetings, inspections, speeches, or appointments were the center of the title; separating policy disclosure from governance and enforcement by distinguishing rule release from rule enforcement; and separating development performance from city-image content by distinguishing projects and economic upgrading from tourism, festivals, and branding. Coders were blind to model output and to all engagement outcomes. Disagreements were adjudicated to produce the final gold-standard label. The 2,000-title audit sample was held out from prompt design and used only for validation.

B.3 LLM Prompt

The full-corpus pass used OpenAI with `model = gpt-5.4` and `reasoning_effort = low` under a deterministic single-label instruction. The admissible Chinese primary labels were 意识形态与宣传教育, 时政与领导活动, 公共服务信息, 社会保障与公共福利, 应急管理与风险沟通, 政策与政务公开, 社会治理与执法通报, 群众动员与社会参与, 经济与发展建设, and 城市形象与文化活动. The production prompt was set before large-scale assignment and used for both the corpus pass and the validation audit. The prompt was:

Prompt used for title-based coding

```
You are a Chinese political-communication annotator.
Task: classify a local-government WeChat post into a fixed ten-label taxonomy.
Use only title-side inputs. Do not use engagement metrics or infer category from popularity.
Choose exactly one primary label according to the post's main communicative purpose.
Decision rule: ask whether the title mainly teaches political values, reports leaders or political
events, provides practical service information, describes welfare or benefit provision, manages
immediate risk, discloses policy or administrative procedure, reports enforcement or governance action,
mobilizes participation, promotes development, or projects culture and city image.
Boundary rules:
- public service information vs. social protection and welfare: procedure and daily-use guidance vs.
welfare substance, health, education, jobs, subsidies, and pensions;
- leadership and political events vs. policy disclosure: leaders, meetings, inspections, speeches, and
appointments vs. policy texts, notices, budgets, and formal disclosure;
- policy disclosure vs. governance and enforcement: rule release vs. rule enforcement;
- emergency and risk communication vs. governance and enforcement: urgent warnings and hazards vs.
routine control and punishment;
- economic development vs. city image: projects and performance claims vs. tourism, festivals,
exhibitions, and branding.
Return JSON only in the following schema:
{"primary_label":"...", "family_label":"...", "confidence":0.00, "brief_rationale":"..."}
Input fields:
title = {title}
public_account_name = {public_account_name}
province = {province}
city = {city}
publish_date = {publish_date}
keywords = {keywords}
reason = {reason}
```

There was no fine-tuning stage and no supervised training split in the machine-learning sense. The discovery sample was used to finalize the codebook and prompt, after which the 2,000-title coder audit was held out for validation.

Table SI.7: Inter-coder reliability for the dual-coded 2,000-title validation sample.

statistic	value
Primary-label exact agreement	0.8875
Family-label exact agreement	0.9130
Primary-label Cohen's kappa	0.8726
Family-label Cohen's kappa	0.8811
Adjudication rate	0.1125

Table SI.8: Overall validation metrics for detailed-label and family-level LLM coding.

task	accuracy	macro_f1	evaluation_titles
Detailed labels	0.8150	0.7955	2,000
Higher-order families	0.8755	0.8642	2,000

B.4 Validation Results

Table SI.7 reports the reliability of the dual-coded audit sample. Exact agreement reaches 0.8875 at the detailed-label level and 0.9130 at the family level. Cohen's κ is 0.8726 for the ten-label task and 0.8811 for the four-family task. Only 11.25% of titles required adjudication, which is consistent with the fact that the hardest disagreements cluster around a small set of expected boundary zones rather than around wholesale coder confusion.

Against the adjudicated gold-standard labels, the LLM reaches 0.8150 accuracy and 0.7955 macro F1 on the ten-label task. At the family level, accuracy rises to 0.8755 and macro F1 to 0.8642. Table SI.8 reports both sets of overall validation metrics together. This higher family-level performance is substantively important because the paper's theory-facing analyses are conducted primarily at the family level.

Table SI.9 shows that family-level performance is strongest for soft propaganda and public service, while the weakest family is state governance, which is also the most semantically intermediate family in the codebook. Table SI.10 shows the same pattern at the ten-label level: performance is strongest for city-image and cultural content, economic development, leadership events, and emergency communication, and weaker for mass mobilization, policy disclosure, welfare content, and general public-service information. Those are also the labels where discovery review and coder adjudication revealed the greatest boundary pressure.

As a complementary face-validity check, Figure SI.2 visualizes family-specific keyword clouds built from the archive's auxiliary keyword field. To keep the clouds informative rather than dominated by generic high-frequency terms, the plotted keywords are weighted by family-level TF-IDF rather than by raw counts alone, after splitting the keyword strings on punctuation delimiters. The resulting semantic neighborhoods line up closely with the intended taxonomy: hard-propaganda titles emphasize political leaders, meetings, and ideological-study language; public-service titles emphasize health, weather, safety, recruitment, and epidemic-response language; state-governance titles emphasize policy release, disclosure, rectification, and participation; and soft-propaganda titles emphasize investment, tourism, culture, projects, and development performance. This fig-

Table SI.9: Class-wise validation metrics for the four higher-order families.

family	precision	recall	f1	support
Hard Propaganda	0.8478	0.8966	0.8715	435
Public Service	0.8959	0.9016	0.8987	630
Soft Propaganda	0.9455	0.8691	0.9057	619
State Governance	0.7567	0.8070	0.7810	316

Table SI.10: Class-wise validation metrics for the ten detailed labels.

detailed_label	family	precision	recall	f1	support
Public Service Information	Public Service	0.6911	0.8019	0.7424	212
City Image and Cultural Activity	Soft Propaganda	0.9280	0.8974	0.9125	273
Emergency and Risk Communication	Public Service	0.8000	0.8848	0.8403	217
Ideological Education	Hard Propaganda	0.7304	0.9313	0.8187	160
Policy and Administrative Disclosure	State Governance	0.6875	0.7857	0.7333	168
Leadership and Political Events	Hard Propaganda	0.8750	0.8145	0.8437	275
Social Protection and Welfare	Public Service	0.8649	0.6368	0.7335	201
Social Governance and Enforcement	State Governance	0.8247	0.8081	0.8163	99
Economic Development and Construction	Soft Propaganda	0.9115	0.8035	0.8541	346
Mass Mobilization and Participation	State Governance	0.6667	0.6531	0.6598	49

ure is not a substitute for the coder audit, but it provides an additional full-corpus check that the higher-order families are semantically coherent.

Figure SI.3 visualizes the family-level confusion matrix. Most remaining error is not a collapse of propaganda and service into one undifferentiated category. Instead, it is concentrated in the expected intermediate zone where state-governance titles can be absorbed into adjacent categories. That pattern is consistent with the codebook logic and helps justify using the four families for diagnostics while reserving the three main families for the paper's core theoretical tests.

C Robustness and Additional Analyses

C.1 Bandwidth Sensitivity

Figure SI.4 plots the robust bias-corrected RDD estimate for one-click engagement rate as a function of bandwidth at each cutoff. The 2018 estimate is stable and negative across all bandwidths from 3 to 25 days. The 2020 estimate is stable and positive. Both are robust to the choice of bandwidth.

Table SI.11 reports a comprehensive set of robustness checks for the one-click engagement rate at both cutoffs: bandwidth sensitivity (0.5x–2x MSE-optimal), polynomial order (local mean, linear, quadratic), donut hole (excluding ±1–3 days), alternative kernels (triangular, uniform, Epanechnikov), and CER-optimal bandwidth.

Table SI.12 runs the same battery for the two 2020 component outcomes. Across every specification, the restored like channel remains positive, while the *zaikan* channel remains negative. For like, the estimated step ranges from 4.2 to 5.9 percentage points; for *zaikan*, it ranges from −1.6 to −0.9 points.

Its function is mechanism robustness. The table shows that the 2020 decomposition is not a fragile byproduct of one preferred bandwidth or polynomial choice: the same sign pattern survives every standard RDD perturbation.

C.2 Alternative Outcome Definitions

Table SI.13 turns each channel into an indicator for any positive engagement, isolating whether the interface changes the probability of leaving any trace at all. In 2018, the probability of any one-click engagement falls by 11.4 percentage points, while any share remains flat. In 2020, any one-click engagement rises by 29.2 points, driven by a 41.1-point jump in any like; any share again

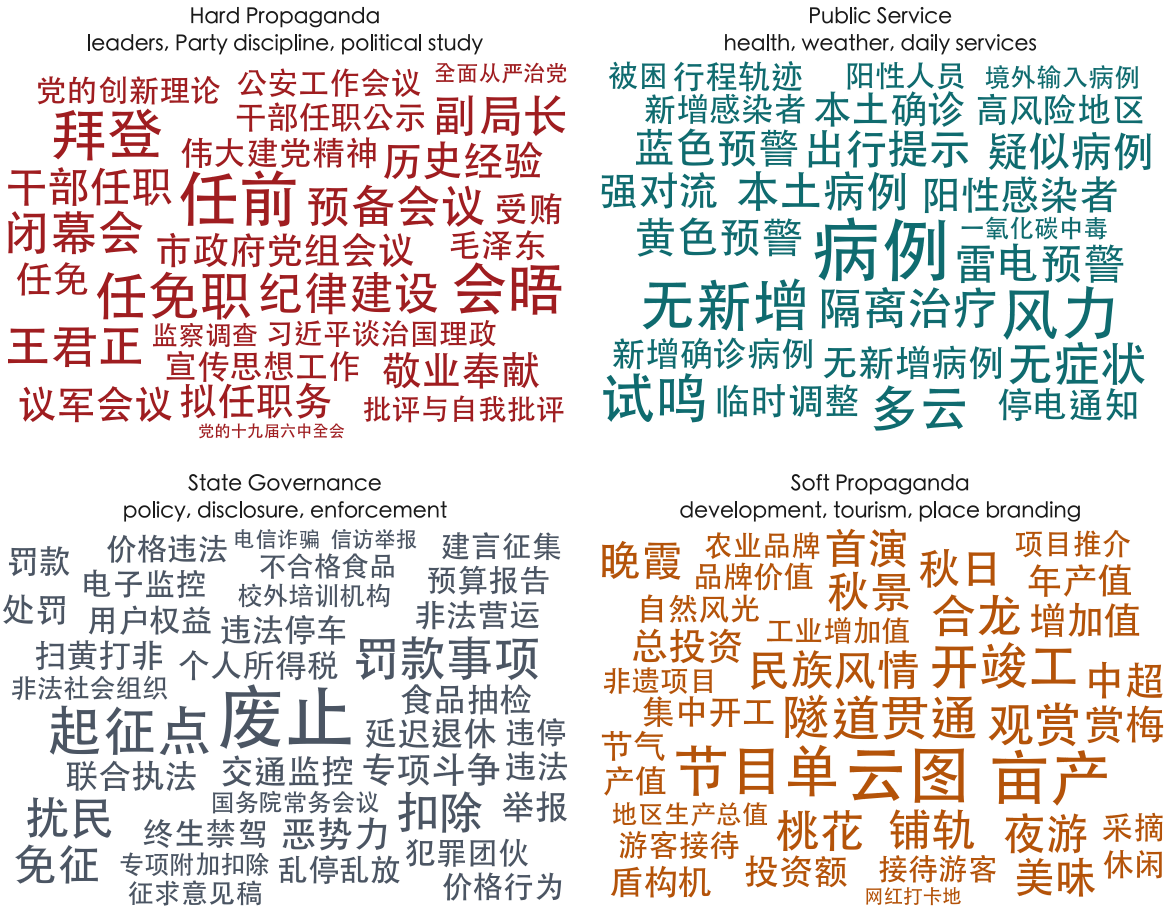


Figure SI.2: Family-Specific Keyword Clouds from the Auxiliary Keyword Field

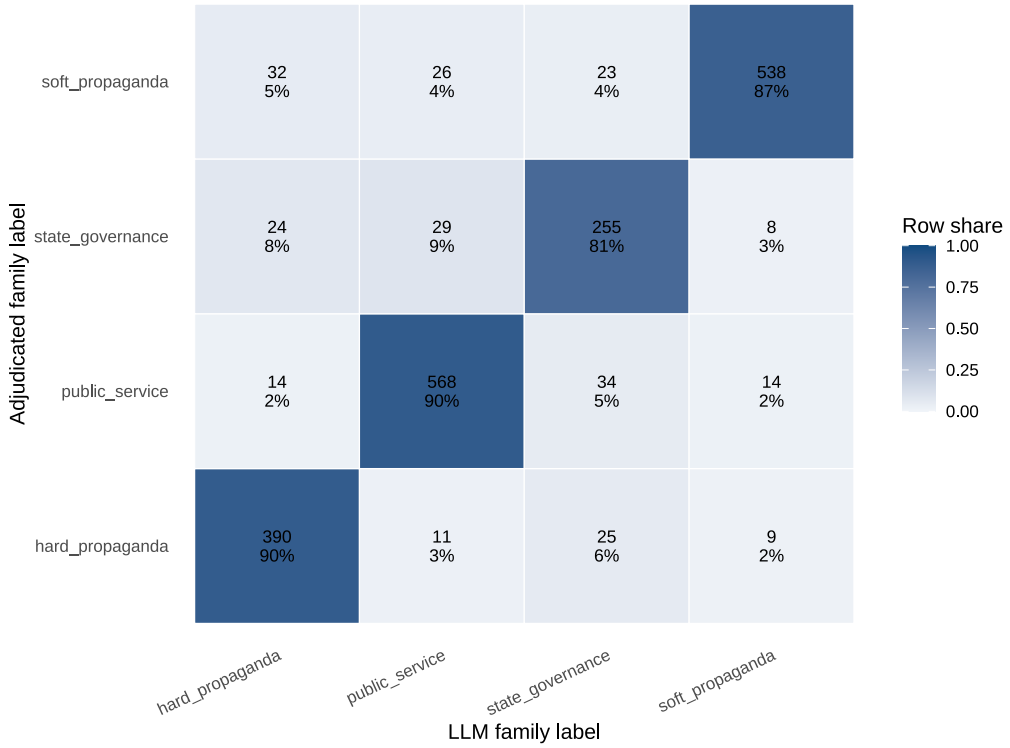


Figure SI.3: Family-Level Confusion Matrix for the WeChat Classification Audit

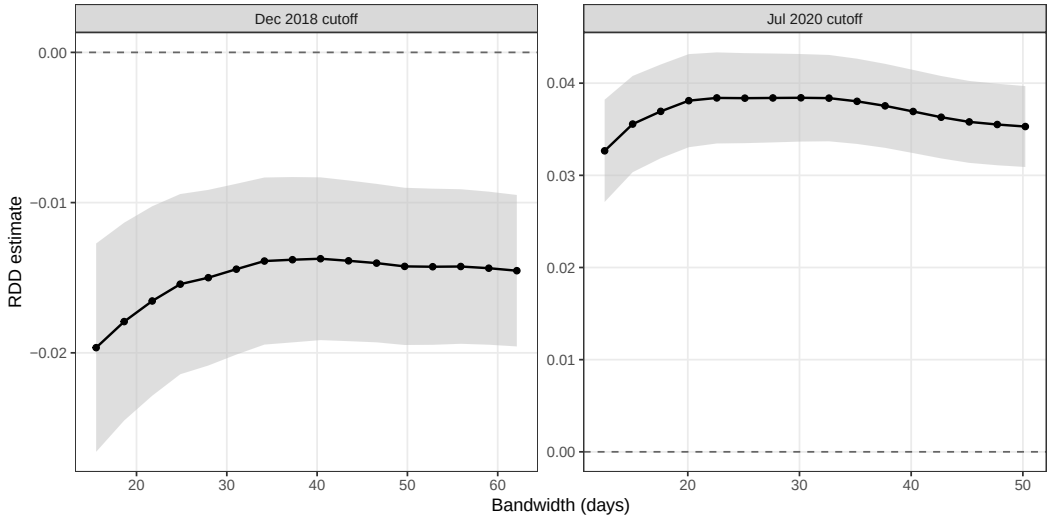


Figure SI.4: Bandwidth sensitivity of the one-click engagement RDD estimate.

Note: Each point is a separate RDD estimate at a given bandwidth. Grey band shows the 95% robust bias-corrected CI. Account-clustered SEs, local linear, triangular kernel.

Table SI.11: Robustness of RDD estimates for one-click rate

Specification	Estimate	SE	p-value	BW	N (L)	N (R)
Baseline (2018)	-0.0136	0.0028	1.302e-06	31	30,652	30,257
BW 0.50x (2018)	-0.0197	0.0035	2.763e-08	16	14,786	14,806
BW 0.75x (2018)	-0.0160	0.0031	3.044e-07	23	22,808	22,200
BW 1.25x (2018)	-0.0138	0.0028	7.864e-07	39	37,365	37,125
BW 1.50x (2018)	-0.0140	0.0027	1.850e-07	47	45,197	44,814
BW 1.75x (2018)	-0.0142	0.0026	6.628e-08	54	52,493	49,855
BW 2.00x (2018)	-0.0145	0.0026	1.612e-08	62	59,790	57,584
Poly p=0 (2018)	-0.0139	0.0026	8.019e-08	24	22,808	22,200
Poly p=1 (2018)	-0.0136	0.0028	1.302e-06	31	30,652	30,257
Poly p=2 (2018)	-0.0147	0.0031	1.880e-06	33	32,341	32,543
Donut +/-1 days (2018)	-0.0108	0.0030	3.465e-04	32	29,459	28,453
Donut +/-2 days (2018)	-0.0082	0.0037	0.026	21	18,218	18,726
Donut +/-3 days (2018)	-0.0096	0.0037	0.009	27	22,574	23,378
Kernel: triangular (2018)	-0.0136	0.0028	1.302e-06	31	30,652	30,257
Kernel: uniform (2018)	-0.0121	0.0028	1.611e-05	22	21,649	21,655
Kernel: epanechnikov (2018)	-0.0131	0.0028	2.680e-06	30	28,404	28,593
CER-optimal (2018)	-0.0137	0.0028	1.245e-06	23	21,649	21,655
Baseline (2020)	0.0378	0.0023	1.282e-58	25	33,879	35,012
BW 0.50x (2020)	0.0327	0.0028	8.797e-31	13	15,866	17,151
BW 0.75x (2020)	0.0374	0.0026	2.018e-47	19	24,215	25,228
BW 1.25x (2020)	0.0384	0.0024	1.828e-57	31	42,925	43,862
BW 1.50x (2020)	0.0375	0.0023	1.305e-58	38	52,739	52,436
BW 1.75x (2020)	0.0360	0.0023	1.431e-56	44	61,766	60,263
BW 2.00x (2020)	0.0353	0.0022	5.688e-56	50	71,934	69,892
Poly p=0 (2020)	0.0381	0.0024	9.796e-59	14	19,103	20,323
Poly p=1 (2020)	0.0378	0.0023	1.282e-58	25	33,879	35,012
Poly p=2 (2020)	0.0380	0.0026	2.075e-47	28	38,742	39,646
Donut +/-1 days (2020)	0.0460	0.0030	7.260e-52	19	22,459	22,304
Donut +/-2 days (2020)	0.0613	0.0044	2.332e-44	12	10,890	11,143
Donut +/-3 days (2020)	0.0527	0.0042	3.028e-36	15	15,996	16,558
Kernel: triangular (2020)	0.0378	0.0023	1.282e-58	25	33,879	35,012
Kernel: uniform (2020)	0.0394	0.0025	1.089e-55	16	20,793	21,930
Kernel: epanechnikov (2020)	0.0366	0.0023	3.467e-57	28	38,742	39,646
CER-optimal (2020)	0.0387	0.0024	4.673e-58	18	24,215	25,228

Table SI.12: Robustness of 2020 RDD estimates for like and zaikan rates

Specification	Estimate	SE	p-value	BW	N (L)	N (R)
Like: Baseline	0.0461	0.0016	6.552e-179	11	14,243	15,643
Like: BW 0.50x	0.0423	0.0020	1.824e-104	6	6,371	7,703
Like: BW 0.75x	0.0426	0.0018	1.864e-123	8	10,831	12,380
Like: BW 1.25x	0.0427	0.0016	3.790e-149	14	17,452	18,699
Like: BW 1.50x	0.0456	0.0016	1.642e-176	17	22,371	23,526
Like: BW 1.75x	0.0480	0.0016	3.042e-193	19	25,837	26,747
Like: BW 2.00x	0.0498	0.0016	2.567e-208	22	30,571	31,612
Like: Poly p=0	0.0512	0.0016	2.091e-237	21	29,008	29,975
Like: Poly p=1	0.0461	0.0016	6.552e-179	11	14,243	15,643
Like: Poly p=2	0.0515	0.0016	1.703e-216	27	37,131	38,040
Like: Donut +/-1 days	0.0551	0.0027	9.397e-91	10	11,525	11,937
Like: Donut +/-2 days	0.0468	0.0047	1.211e-23	8	5,811	6,311
Like: Donut +/-3 days	0.0592	0.0082	6.432e-13	7	4,367	5,439
Like: Kernel: triangular	0.0461	0.0016	6.552e-179	11	14,243	15,643
Like: Kernel: uniform	0.0541	0.0017	1.611e-227	14	17,452	18,699
Like: Kernel: epanechnikov	0.0442	0.0016	1.510e-162	9	10,831	12,380
Like: CER-optimal	0.0433	0.0016	3.060e-156	8	10,831	12,380
Zaikan: Baseline	-0.0120	0.0015	3.220e-16	27	35,546	36,509
Zaikan: BW 0.50x	-0.0085	0.0019	1.170e-05	13	17,452	18,699
Zaikan: BW 0.75x	-0.0106	0.0017	1.697e-10	20	27,439	28,334
Zaikan: BW 1.25x	-0.0119	0.0015	4.402e-16	33	45,602	46,201
Zaikan: BW 1.50x	-0.0113	0.0014	6.401e-15	40	56,840	55,552
Zaikan: BW 1.75x	-0.0113	0.0014	2.808e-15	47	65,337	63,604
Zaikan: BW 2.00x	-0.0116	0.0014	7.534e-17	54	76,010	73,231
Zaikan: Poly p=0	-0.0130	0.0014	7.283e-20	11	14,243	15,643
Zaikan: Poly p=1	-0.0120	0.0015	3.220e-16	27	35,546	36,509
Zaikan: Poly p=2	-0.0119	0.0017	2.547e-12	26	33,879	35,012
Zaikan: Donut +/-1 days	-0.0131	0.0018	6.797e-13	21	27,252	27,051
Zaikan: Donut +/-2 days	-0.0156	0.0024	1.159e-10	16	19,018	19,026
Zaikan: Donut +/-3 days	-0.0139	0.0048	0.004	10	8,484	9,489
Zaikan: Kernel: triangular	-0.0120	0.0015	3.220e-16	27	35,546	36,509
Zaikan: Kernel: uniform	-0.0124	0.0015	6.329e-17	17	22,371	23,526
Zaikan: Kernel: epanechnikov	-0.0132	0.0015	1.095e-18	22	30,571	31,612
Zaikan: CER-optimal	-0.0126	0.0015	3.483e-17	19	25,837	26,747

Table SI.13: RDD estimates for extensive-margin engagement outcomes. Each outcome equals one when the post receives any positive count on that engagement channel.

Specification	Estimate	SE	p-value	BW	N (L)	N (R)
2018: Any one-click	-0.1135	0.0264	1.733e-05	5	5,134	5,773
2018: Any share	0.0050	0.0148	0.736	13	11,918	11,990
2020: Any one-click	0.2917	0.0195	1.048e-50	6	7,375	9,211
2020: Any like	0.4114	0.0125	1.787e-237	5	6,371	7,703
2020: Any zaikan	0.0605	0.0165	2.393e-04	8	10,831	12,380
2020: Any share	0.0032	0.0188	0.865	7	9,164	10,811

Table SI.14: Placebo cutoff tests for one-click rate

Specification	Estimate	SE	p-value	BW	N (L)	N (R)
2018-06-21	-0.0011	0.0038	0.779	11	10,017	11,361
2018-09-21	-0.0012	0.0050	0.813	7	5,821	6,073
2019-03-21	0.0018	0.0038	0.633	6	5,786	7,050
2019-06-21	-0.0064	0.0032	0.045	9	9,468	9,635

remains flat. Any *zaikan* also rises modestly, even though the average *zaikan* rate falls in the main specifications.

Its function is participation-margin decomposition. The table separates the decision to leave any trace from the amount of engagement conditional on participation, and it shows that the 2020 redesign mainly widened participation in low-visibility approval without generating a parallel increase in sharing.

C.3 Placebo Cutoffs

Table SI.14 tests the one-click rate at four non-event dates spaced ±3 and ±6 months from the December 2018 cutoff. Three of four placebos are not significant. The March 2019 date ($p = 0.035$) falls near the haokan-to-zaikan rename, consistent with a small behavioral effect of the wording change.

The function of this table is to show that the main 2018 result is not easy to reproduce at arbitrary nearby dates. That matters because it pushes against purely seasonal or slowly evolving explanations.

Table SI.15 formally tests the March 15, 2019 rename as a placebo: the one-click rate shows a modest positive shift ($p = 0.045$), while share and reads are unaffected. This supports the interpretation that the rename slightly reduced social pressure but did not constitute a fundamental visibility change.

Its function is interpretive cleanup. The table distinguishes a modest wording effect from the much larger redesign effects that altered the visibility structure of engagement.

Table SI.15: Placebo RDD at 2019-03-15 Haokan-to-Zaikan rename

Specification	Estimate	SE	p-value	BW	N (L)	N (R)
One-click rate	0.0070	0.0035	0.045	9	9,172	9,282
Share rate	-0.0001	0.0007	0.892	8	8,024	8,763
Reads	194.7956	174.6538	0.265	18	17,312	17,710

Table SI.16: Contrast test at the May 2017 discovery reform. This change broadened content discovery without directly rebundling article-footer approval buttons.

Specification	Estimate	SE	p-value	BW	N (L)	N (R)
Reads	-227.9468	185.8091	0.220	16	13,013	12,847
One-click rate	0.0125	0.0046	0.007	5	3,636	4,647
Share rate	-0.0008	0.0007	0.253	9	6,550	7,459

Table SI.17: Account-day aggregated RDD estimates. Each account-day receives equal weight, reducing the influence of high-volume posting accounts.

Specification	Estimate	SE	p-value	BW	N (L)	N (R)
Account-day (2018): One-click	-0.0185	0.0040	2.792e-06	8	1,327	1,549
Account-day (2020): One-click	0.0255	0.0039	4.893e-11	4	917	1,064
Account-day (2020): Like	0.0398	0.0018	3.135e-110	6	1,094	1,323
Account-day (2020): Zaikan	-0.0115	0.0025	4.787e-06	6	1,287	1,576
Account-day (2020): Share	0.0002	0.0006	0.745	12	2,395	2,665

C.4 2017 Discovery Reform as a Contrast Case

Table SI.16 treats the May 2017 discovery reform as a contrast rather than as a third main visibility design. Reads and share remain continuous at that cutoff, while one-click engagement increases modestly. This pattern is consistent with the idea that 2017 altered distribution and discovery conditions without directly recreating the expressive rebundling logic that drives the 2018 and 2020 results.

The function of this contrast table is conceptual. It shows that not every WeChat interface reform belongs in the same theoretical bucket: discovery changes look different from approval-visibility changes.

C.5 Account-Day Aggregation

Table SI.17 re-estimates the main designs after collapsing the data to the account-day level. Each account-day receives equal weight, reducing the influence of prolific posting accounts. The sign pattern and magnitudes remain close to the article-level results.

Its function is weighting robustness. The table addresses the concern that article-level estimates might be driven disproportionately by a small number of high-volume accounts.

C.6 Alternative 2020 Samples

Table SI.18 shows that the 2020 unbundling results are not driven solely by COVID-era emergency communication or by the most politically salient outbreak region. Excluding emergency-category and COVID-linked posts leaves the estimates essentially unchanged, and excluding Hubei and Wuhan produces the same qualitative pattern.

Its function is alternative-explanation control. The table directly addresses the possibility that the 2020 result is really a pandemic-composition artifact rather than an interface effect.

C.7 Extended Balance and Composition Diagnostics

Table SI.19 expands the core balance tests to title length, head-article share, all four content families, and emergency/COVID-linked content shares. Some of these diagnostics show modest local

Table SI.18: Alternative-sample checks for the July 2020 cutoff. The first panel excludes emergency-category and COVID-linked posts; the second excludes Hubei and Wuhan accounts.

Specification	Estimate	SE	p-value	BW	N (L)	N (R)
No COVID/emergency: One-click	0.0288	0.0040	4.226e-13	5	4,372	4,985
No COVID/emergency: Like	0.0431	0.0019	5.414e-112	7	5,698	7,419
No COVID/emergency: Zaikan	-0.0134	0.0024	2.009e-08	7	5,698	7,419
Exclude Hubei/Wuhan: One-click	0.0270	0.0042	8.385e-11	4	4,595	5,162
Exclude Hubei/Wuhan: Like	0.0429	0.0019	1.813e-116	7	7,080	8,862
Exclude Hubei/Wuhan: Zaikan	-0.0121	0.0024	2.906e-07	6	7,080	8,862

Table SI.19: Extended balance and composition diagnostics at each cutoff. Account-clustered SEs.

Specification	Estimate	SE	p-value	BW	N (L)	N (R)
Reads (2018)	245.1400	207.7260	0.238	9	9,110	9,794
Reads (2020)	212.6953	270.2765	0.431	8	10,831	12,380
Title length (2018)	-0.1980	0.4288	0.644	11	11,398	10,873
Title length (2020)	-1.2588	0.5460	0.021	6	6,371	7,703
Head article share (2018)	0.0207	0.0092	0.024	9	9,110	9,794
Head article share (2020)	0.0324	0.0111	0.004	6	6,371	7,703
Hard prop. (2018)	-0.0387	0.0206	0.060	9	9,110	9,794
Hard prop. (2020)	0.0113	0.0221	0.609	7	9,164	10,811
Soft prop. (2018)	0.0722	0.0236	0.002	7	5,706	6,945
Soft prop. (2020)	0.0171	0.0205	0.402	6	7,375	9,211
State gov. (2018)	-0.0645	0.0243	0.008	6	5,134	5,773
State gov. (2020)	-0.0132	0.0190	0.487	5	6,371	7,703
Pub. service (2018)	-0.0234	0.0247	0.342	5	4,623	4,624
Pub. service (2020)	0.0152	0.0234	0.517	6	7,375	9,211
Emergency/risk share (2018)	0.0133	0.0087	0.127	6	5,706	6,945
Emergency/risk share (2020)	0.0348	0.0139	0.012	6	7,375	9,211
COVID keyword share (2018)	-0.0052	0.0043	0.229	6	5,706	6,945
COVID keyword share (2020)	0.0394	0.0130	0.003	6	7,375	9,211

composition shifts, which is precisely why the paper keeps the main-text balance table narrow and theory-linked. The central one-click estimates nevertheless remain stable in the alternative-sample and account-day checks above.

The function of the extended balance table is diagnostic transparency. Rather than hiding marginal composition changes, it shows exactly where they appear and why the paper leans on a bundle of robustness checks rather than a single balance screen.

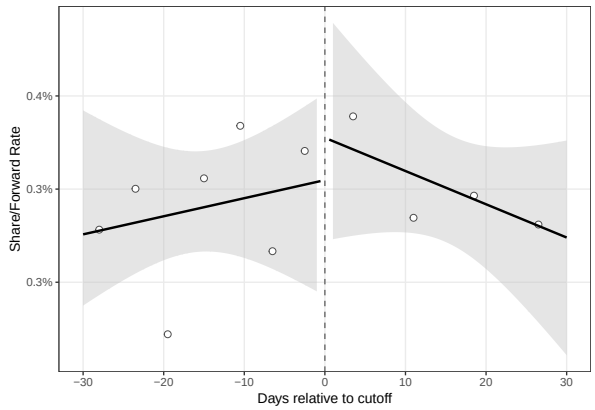
C.8 Additional Outcome Clarification Plots

Figure SI.5 collects outcome-specific RDD plots that sharpen the mechanism discussion in the main text. The 2018 share plot shows no visible step, while the 2020 panels show recovery in one-click engagement, a strong increase in the restored like channel, and little change in sharing.

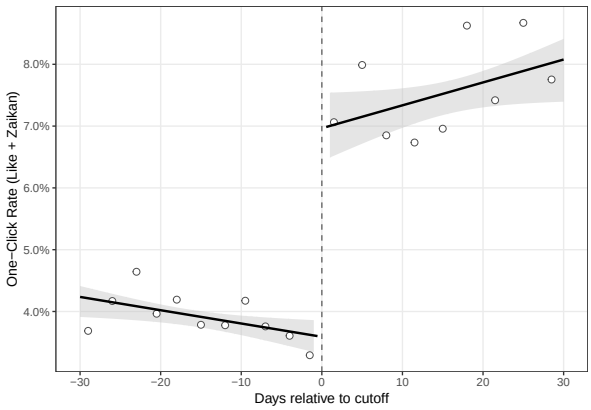
Each panel has a distinct job: the 2018 share panel functions as a placebo visual, the 2020 one-click panel shows the aggregate recovery, the 2020 like panel isolates the restored low-visibility channel, and the 2020 share panel shows that broader forwarding behavior does not move in parallel.

C.9 Category-Level Heterogeneity

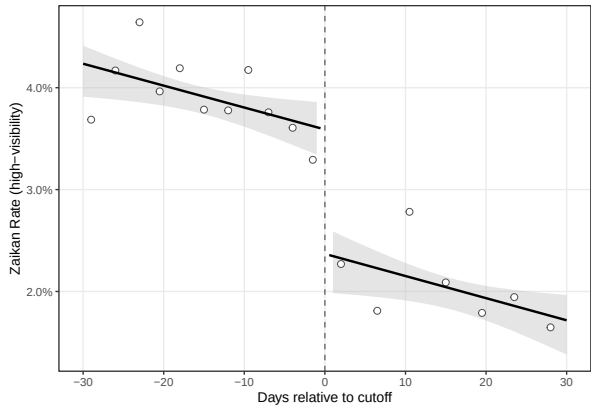
Table SI.20 extends the H4 heterogeneity analysis from four families to all ten content categories, providing finer-grained evidence for the political-risk gradient.



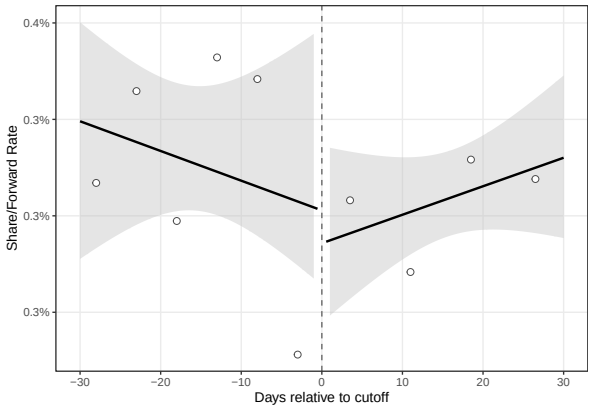
(a) 2018 share rate



(b) 2020 one-click rate



(c) 2020 like rate



(d) 2020 share rate

Figure SI.5: Additional RDD plots for placebo and clarification outcomes.

Note: Panels report daily-aggregated binned means with local linear fits and 95% confidence intervals. Together they show that the main discontinuities are concentrated in the one-click approval channel rather than in sharing.

Table SI.20: RDD estimates at the July 2020 cutoff by content category.

Category	Specification	Estimate	SE	p-value	BW	N (L)	N (R)
Public Service Info	Public Service Info: One-click	0.0443	0.0061	4.596e-13	12	2,014	2,791
Public Service Info	Public Service Info: Like	0.0450	0.0040	9.712e-30	7	993	1,731
Public Service Info	Public Service Info: Zaikan	-0.0070	0.0062	0.257	8	1,265	1,989
City Image & Culture	City Image & Culture: One-click	0.0354	0.0117	0.003	6	475	450
City Image & Culture	City Image & Culture: Like	0.0482	0.0075	1.090e-10	6	475	450
City Image & Culture	City Image & Culture: Zaikan	-0.0089	0.0073	0.225	8	886	675
Emergency & Risk	Emergency & Risk: One-click	0.0384	0.0079	1.282e-06	6	1,252	1,271
Emergency & Risk	Emergency & Risk: Like	0.0431	0.0042	1.069e-24	7	1,462	1,515
Emergency & Risk	Emergency & Risk: Zaikan	0.0010	0.0041	0.803	10	2,608	2,501
Ideological Education	Ideological Education: One-click	0.0105	0.0082	0.202	5	672	892
Ideological Education	Ideological Education: Like	0.0422	0.0035	3.751e-34	8	1,037	1,175
Ideological Education	Ideological Education: Zaikan	-0.0210	0.0042	4.962e-07	10	1,347	1,436
Policy & Disclosure	Policy & Disclosure: One-click	0.0248	0.0059	2.544e-05	6	668	763
Policy & Disclosure	Policy & Disclosure: Like	0.0316	0.0034	1.383e-20	6	668	763
Policy & Disclosure	Policy & Disclosure: Zaikan	-0.0074	0.0042	0.081	7	729	900
Leadership & Political Events	Leadership & Political Events: One-click	0.0402	0.0085	2.021e-06	6	612	752
Leadership & Political Events	Leadership & Political Events: Like	0.0538	0.0061	1.878e-18	6	742	928
Leadership & Political Events	Leadership & Political Events: Zaikan	-0.0131	0.0040	0.001	8	981	1,135
Welfare & Social Protection	Welfare & Social Protection: One-click	0.0422	0.0091	4.074e-06	8	582	809
Welfare & Social Protection	Welfare & Social Protection: Like	0.0412	0.0047	2.018e-18	7	501	657
Welfare & Social Protection	Welfare & Social Protection: Zaikan	0.0040	0.0061	0.508	9	702	956
Governance & Enforcement	Governance & Enforcement: One-click	0.0216	0.0110	0.050	6	318	360
Governance & Enforcement	Governance & Enforcement: Like	0.0358	0.0072	7.331e-07	6	350	420
Governance & Enforcement	Governance & Enforcement: Zaikan	-0.0125	0.0055	0.023	13	834	933
Economic Development	Economic Development: One-click	0.0304	0.0094	0.001	6	1,026	1,274
Economic Development	Economic Development: Like	0.0420	0.0043	1.021e-22	6	1,026	1,274
Economic Development	Economic Development: Zaikan	-0.0134	0.0050	0.008	10	1,908	2,180
Mobilization & Participation	Mobilization & Participation: One-click	0.0306	0.0130	0.019	10	241	351
Mobilization & Participation	Mobilization & Participation: Like	0.0396	0.0092	1.776e-05	9	241	351
Mobilization & Participation	Mobilization & Participation: Zaikan	-0.0101	0.0088	0.250	10	269	376

Its function is granularity. The table shows that the family-level gradient is not an artifact of broad category aggregation but survives when the analysis moves to the underlying ten-category taxonomy.

C.10 Family-Specific 2018 RDD Plots

Figure SI.6 reports the 2018 one-click discontinuity separately by content family. The negative step is visible across all four families, complementing the main-text 2020 unbundling figure.

The figure's function is coverage. It shows that the 2018 visibility penalty is not confined to one substantive domain, even though the later 2020 reallocation differs by political risk.

C.11 Family-Specific 2020 One-Click RDD Plots

Figure SI.7 reports the same 2020 cutoff separately by family using the total one-click rate rather than the like/zaikan decomposition. Together with the main-text unbundling figure, these panels show that the overall recovery in one-click engagement is broad-based across families, even though the composition of that recovery differs sharply by political risk.

Its function is bridge-building between aggregate and decomposed evidence. The figure shows that total one-click recovery is broad, while the main-text family unbundling figure explains why that broad recovery does not imply identical underlying expressive behavior across families.

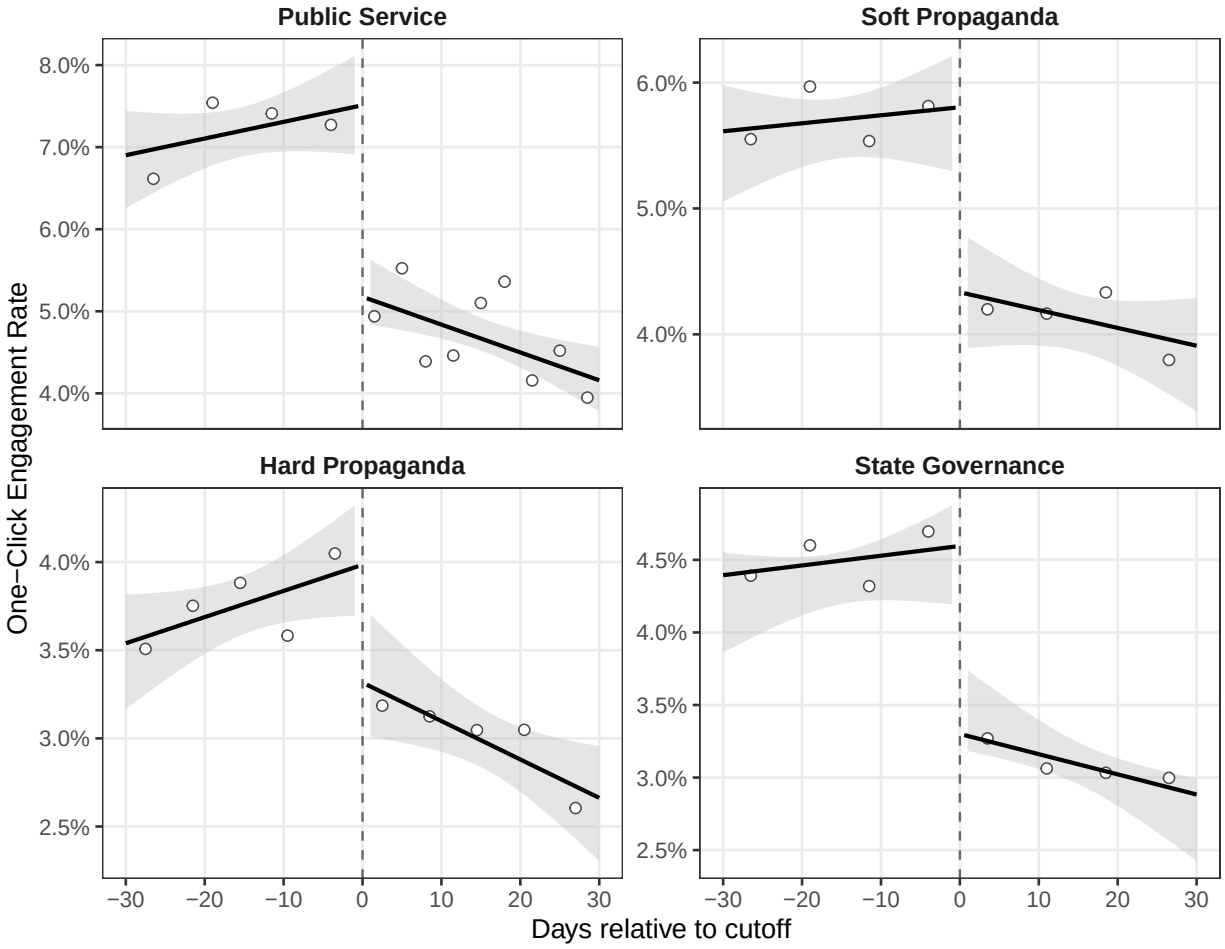


Figure SI.6: One-click engagement rate at the December 2018 cutoff, by content family.

Note: Each panel shows daily-aggregated binned means, local linear fits, and 95% confidence intervals for one-click engagement around the 2018 cutoff.

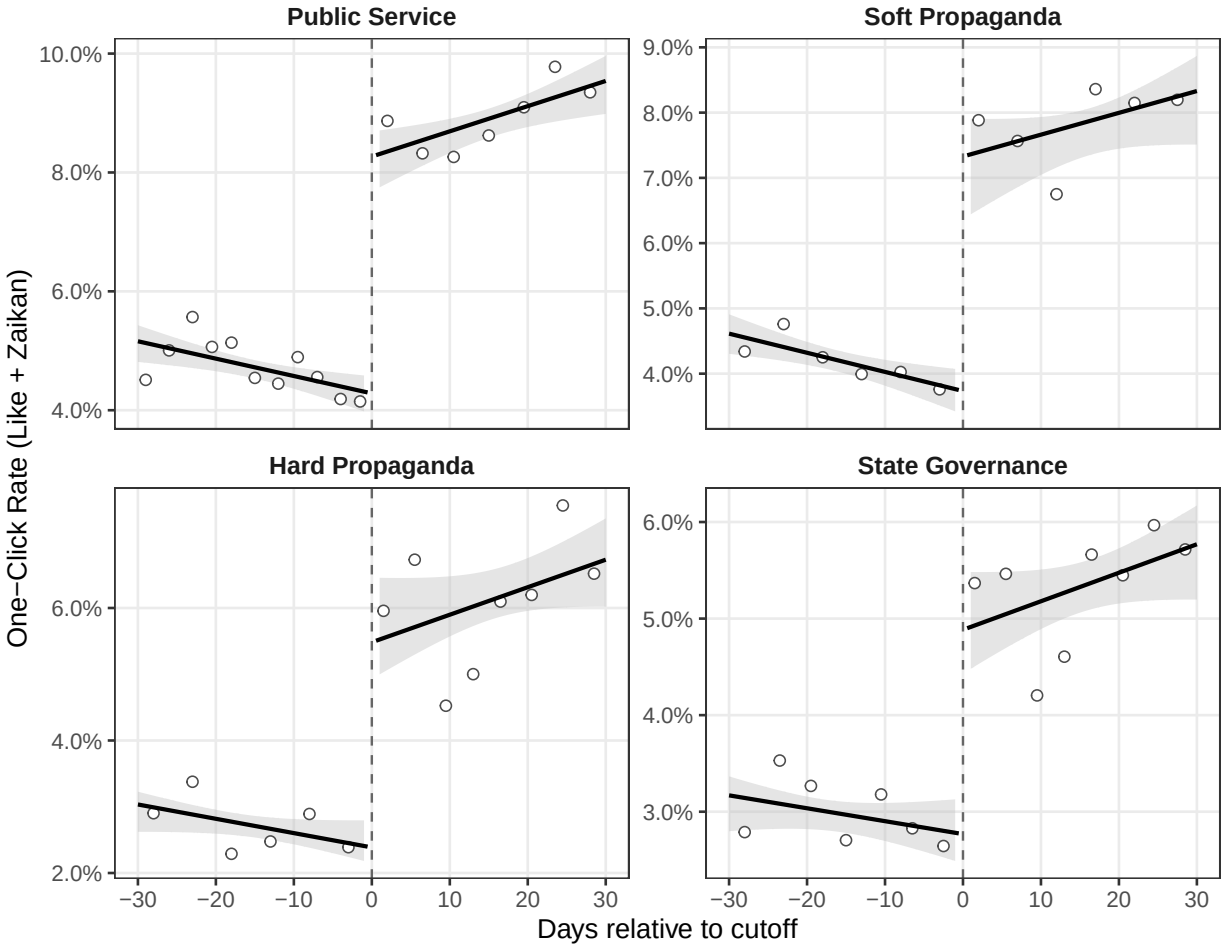


Figure SI.7: One-click engagement rate at the July 2020 cutoff, by content family.

Note: Each panel shows daily-aggregated binned means, local linear fits, and 95% confidence intervals for one-click engagement around the 2020 cutoff.

Table SI.21: RDD estimates at the July 2020 cutoff: head vs.\{ } non-head articles.

Specification	Estimate	SE	p-value	BW	N (L)	N (R)
Head article: One-click	0.0185	0.0096	0.053	4	917	1,064
Head article: Like	0.0384	0.0036	1.453e-26	6	1,094	1,323
Head article: Zaikan	-0.0097	0.0044	0.027	11	2,221	2,514
Non-head article: One-click	0.0325	0.0040	8.160e-16	5	5,277	6,380
Non-head article: Like	0.0439	0.0021	3.100e-101	7	6,088	7,635
Non-head article: Zaikan	-0.0107	0.0026	3.260e-05	6	6,088	7,635

Table SI.22: RDD estimates at the July 2020 cutoff by city size (median split on total posts).

Specification	Estimate	SE	p-value	BW	N (L)	N (R)
Large city: One-click	0.0333	0.0043	6.864e-15	5	5,128	6,157
Large city: Like	0.0462	0.0021	3.163e-104	7	5,963	7,317
Large city: Zaikan	-0.0100	0.0024	2.330e-05	9	8,611	9,734
Small city: One-click	0.0196	0.0058	7.328e-04	5	1,243	1,546
Small city: Like	0.0327	0.0027	4.171e-33	7	1,412	1,894
Small city: Zaikan	-0.0099	0.0042	0.018	7	1,412	1,894

C.12 Head Article versus Non-Head Articles

Table SI.21 compares the RDD effects for head articles (the first article in a push notification, which receives the most prominent placement) versus non-head articles. Non-head articles show larger one-click discontinuities at both cutoffs, consistent with the interpretation that lower-attention articles are more sensitive to interface-design changes.

The function of this table is prominence heterogeneity. It asks whether interface effects depend on article placement and therefore on the amount of ambient attention an article already receives.

C.13 City Size Heterogeneity

Table SI.22 splits the sample at the median city posting volume. Large cities show larger like-rate recovery at 2020 (+0.046 vs. +0.033), consistent with the theoretical expectation that denser social networks amplify the visibility cost of engagement.

Its function is contextual heterogeneity. The table probes whether visibility costs are stronger in communicative environments where social exposure is plausibly denser.

C.14 High-Confidence Classification Subsample

Table SI.23 re-estimates the main RDD specifications using only articles classified with above-median LLM confidence. All estimates remain significant and similar in magnitude to the full-sample results.

The function of this table is measurement robustness. It shows that the main results do not depend on the most ambiguous title classifications.

Table SI.23: RDD estimates at the July 2020 cutoff for above-median classification confidence.

Specification	Estimate	SE	p-value	BW	N (L)	N (R)
High confidence: One-click	0.0286	0.0040	4.914e-13	4	5,465	6,020
High confidence: Like	0.0429	0.0018	1.923e-120	6	7,247	8,963
High confidence: Zaikan	-0.0121	0.0024	4.384e-07	6	7,247	8,963

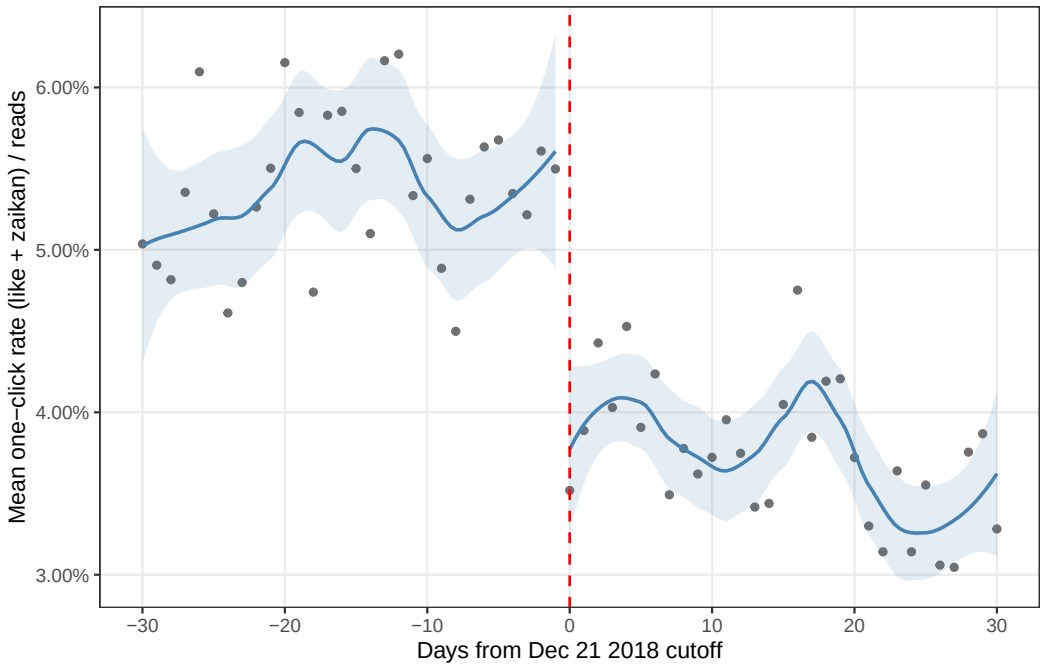


Figure SI.8: Daily one-click engagement rate around the December 2018 cutoff.

Note: Each point is the daily mean one-click rate. Smooth curves are loess fits (span = 0.5) with 95% CIs, fitted separately on each side of the cutoff.

C.15 Event-Study Visualization

Figures SI.8–SI.9 plot raw daily means of the one-click engagement rate around each cutoff, overlaid with loess smoothers fitted separately on each side. These complement the formal RDD estimates by showing the underlying daily variation.

Their function is visual transparency. Instead of relying only on local-polynomial summaries, the figures show the noisier day-level data that the RDD is summarizing.

C.16 Additional Descriptive Material

Table SI.24 reports account-level summary statistics for the 329 prefecture-level government accounts.

Its function is unit-level context: the table clarifies what kinds of accounts populate the archive before article-level estimates aggregate across them.

Table SI.25 reports category-level descriptive statistics for all ten content categories.

Its function is taxonomy resolution. The table lets readers inspect the fine-grained categories that underlie the paper's broader family groupings.

Table SI.26 decomposes the extensive and intensive margins by both interface regime and content family.

Its function is to combine the paper's two main descriptive dimensions: margin of engagement and content type. It shows how participation versus intensity varies not only across regimes but also across substantive families.

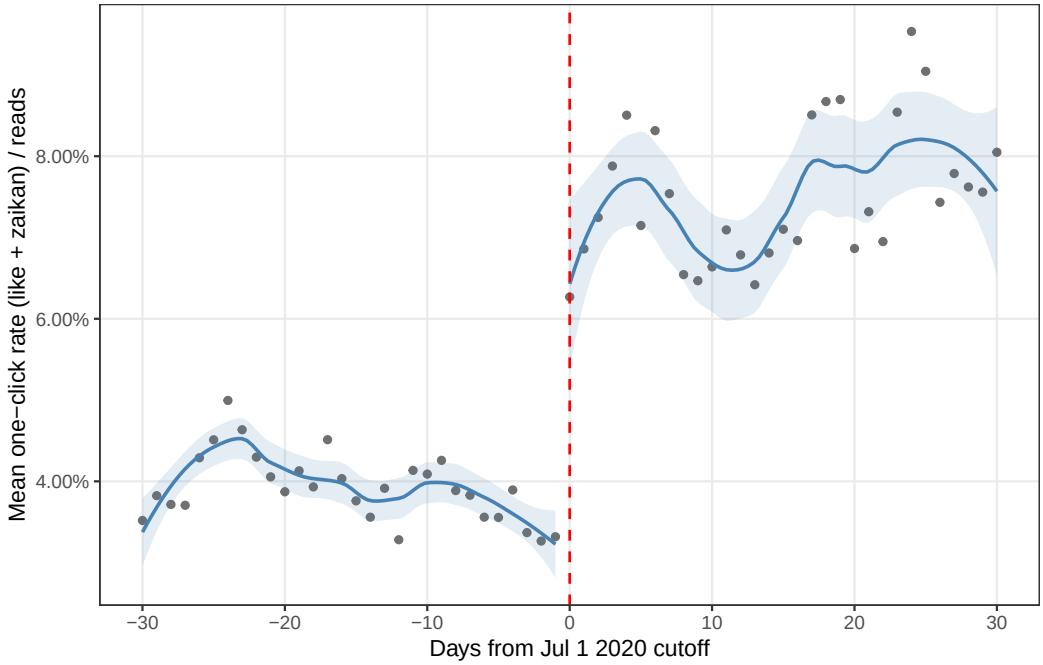


Figure SI.9: Daily one-click engagement rate around the July 2020 cutoff.

Note: Each point is the daily mean one-click rate. Smooth curves are loess fits (span = 0.5) with 95% CIs, fitted separately on each side of the cutoff.

Table SI.24: Account-level summary statistics for the 329 prefecture-level government WeChat accounts.

Statistic	Value
Accounts	329
Posts per account: mean	12,787
Posts per account: median	11,485
Posts per account: SD	8,902
Posts per account: min	169
Posts per account: max	52,380
Mean reads per account: mean	1,620
Mean reads per account: SD	1,563

Table SI.25: Category-level descriptive statistics.

Category	Family	Posts (M)	Reads M (SD)	Likes M (SD)	Shares M (SD)	Zaikan M (SD)
	Public Service	0.52	3,987 (7,392)	268 (770)	15 (54)	143 (465)
	State Governance	7.40e-05	4,500 (8,013)	280 (728)	5 (17)	88 (198)
	Public Service	0.54	3,835 (7,961)	242 (796)	17 (65)	162 (536)
	State Governance	6.80e-05	2,901 (6,006)	129 (448)	4 (8)	63 (264)
	Public Service	0.27	2,149 (4,722)	133 (474)	8 (34)	74 (292)
	State Governance	2.80e-05	1,348 (1,767)	102 (166)	3 (7)	23 (43)
	Soft Propaganda	0.49	2,352 (4,977)	162 (577)	4 (21)	50 (229)
	Soft Propaganda	0.64	1,536 (3,619)	99 (419)	3 (15)	36 (175)
	State Governance	0.20	1,489 (3,663)	63 (277)	2 (11)	22 (134)
	State Governance	0.10	1,252 (3,245)	56 (241)	2 (10)	18 (102)
	State Governance	0.43	1,213 (3,379)	55 (271)	2 (10)	19 (112)
	Hard Propaganda	0.53	1,184 (3,221)	73 (365)	2 (9)	9 (77)
	State Governance	1.25e-04	638 (1,234)	39 (117)	1 (4)	0 (0)
	Hard Propaganda	0.47	875 (2,437)	44 (238)	1 (6)	8 (72)
	State Governance	1.85e-04	656 (1,048)	46 (129)	1 (2)	0 (0)

Table SI.26: Extensive and intensive margins of engagement by interface regime and content family. All values in percent.

Regime	Family	Like Ext.	Like Int.	Zaikan Ext.	Zaikan Int.	Share Ext.	Share Int.	N
Pre-reform (before Dec 2018)	Public Service	49.7	14.0			25.0	1.4	295,417
Pre-reform (before Dec 2018)	Soft Propaganda	38.9	13.8			17.1	1.4	317,633
Pre-reform (before Dec 2018)	State Governance	36.9	11.1			16.7	1.6	197,792
Pre-reform (before Dec 2018)	Hard Propaganda	36.1	9.6			15.5	1.9	211,179
Haokan/Zaikan only (Dec 2018 – Jun 2020)	Public Service			41.9	11.1	32.9	1.3	233,716
Haokan/Zaikan only (Dec 2018 – Jun 2020)	Soft Propaganda			31.7	13.3	22.0	1.3	156,747
Haokan/Zaikan only (Dec 2018 – Jun 2020)	State Governance			30.2	9.9	21.1	1.4	122,726
Haokan/Zaikan only (Dec 2018 – Jun 2020)	Hard Propaganda			29.8	9.4	20.0	1.6	154,719
Like + Zaikan + Share (Jul 2020 –)	Public Service	54.5	10.6	49.4	7.2	37.3	1.2	813,567
Like + Zaikan + Share (Jul 2020 –)	Soft Propaganda	40.2	15.3	37.5	5.2	24.0	1.2	656,264
Like + Zaikan + Share (Jul 2020 –)	State Governance	38.2	11.1	34.9	3.8	23.4	1.5	412,199
Like + Zaikan + Share (Jul 2020 –)	Hard Propaganda	37.4	15.2	34.7	1.9	21.4	1.6	635,113

Figure SI.10 plots the monthly extensive margin (share of posts with any positive engagement) for each metric.

The figure's function is dynamic corroboration. It shows in time-series form that the participation margin moves in the same directions highlighted by the descriptive regime table.

Figure SI.10: Monthly extensive margin of engagement by interface regime.

Note: Share of posts with any positive count for each engagement metric. Gaps and regime annotations follow the same conventions as Figure 3.